

CNN-Based Computer Vision Approach for Local Bangladeshi Colored Flower Recognition with Mobile App Deployment

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**A Capstone project report submitted in partial fulfillment of the
requirements for the degree of
Bachelor of Science in Computer Science and Engineering**



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Declaration

We, **Kazi Minhajul Goni Sami, Md. Sajjad Hossain, Riya Saha, and Tuly Rahman** hereby, declare that the work presented in this capstone project report is the outcome of the investigation performed by us under the supervision of **Dr. Md. Hasanul Ferdous**, Assistant Professor, Department of Computer Science and Engineering, East West University. We also declare that no part of this project has been or is being submitted elsewhere for the award of any degree or diploma, except for publication.

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Letter of Acceptance

The capstone project report entitled “**CNN-Based Computer Vision Approach for Local Bangladeshi Colored Flower Recognition with Mobile App Deployment**” is submitted by **Kazi Minhajul Goni Sami** (2021-2-60-020), **Md. Sajjad Hossain** (2021-2-60-136), **Riya Saha** (2021-2-60-116), and **Tuly Rahman** (2021-1-60-106) to the Department of Computer Science and Engineering, East West University, Dhaka, Bangladesh. The requirement for the Bachelor of Science in Computer Science and Engineering is partially satisfied by this report.

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- I. MobileNetV2**
- II. EfficientNetV2B0**
- III. ResNet50V2**
- IV. InceptionV3**
- V. Xception**

Chapter 1

Introduction

In recent years, the field of computer vision has made significant strides, enabling automation in various domains, including image classification and object recognition. Convolutional Neural Networks (CNNs), in particular, have become a cornerstone for many of these advancements, offering remarkable accuracy in tasks like image classification, pattern recognition, and feature extraction. One of the exciting applications of CNNs is in the field of plant identification, where they help classify and identify species from images, making it easier to understand biodiversity and enhance environmental education. This thesis focuses on the development of a **CNN-Based Computer Vision Approach for Local Bangladeshi Colored Flower Recognition with Web App Deployment**, aiming to automate flower identification using CNN models, specifically tailored to recognize the vibrant and diverse flowers of Bangladesh.

The use of CNNs for flower species recognition has been extensively explored in previous research. A study titled “**Flower Species Identification Using Deep Convolutional Neural Networks**” investigates several deep CNN models, including AlexNet, CaffeNet, and GoogLeNet. The study found that GoogLeNet achieved the highest accuracy, with 67.45% at rank 1 and 90.82% at rank 10, demonstrating the robustness of CNN models in classifying a wide variety of flower species from the PlantCLEF 2015 dataset. This highlights the capacity of CNNs to handle complex image data and offers a benchmark for flower classification using modern deep learning models [1]. Additionally, the study contrasts CNN methods with traditional hand-designed feature extraction techniques like Kernel Descriptors (KDES), SIFT, HOG, and HSV, underlining the superior performance of CNNs in dealing with large-scale and varied datasets.

Building on this, another paper titled “**Flower Species Recognition System Using Convolution Neural Networks and Transfer Learning**” uses transfer learning techniques to improve classification accuracy. By applying Inception-v3, the study achieved an impressive 93.41% accuracy on the FLOWERS102 dataset. This paper emphasizes the potential of transfer learning to boost the performance of CNNs, especially when dealing with datasets with limited samples or when aiming to apply the model to mobile devices, making it highly relevant for the development of mobile and web-based flower recognition apps [2].

Similarly, the paper “**Recognition of Flowers Using Convolutional Neural Networks**” discusses a mobile application designed to identify flowers using deep learning. The authors report an accuracy of 83.13% in classifying four common flower types—Sunflower, Dandelion, Rose, and Tulip. This research highlights the practical application of CNNs for real-time, on-the-go flower identification, proving that CNNs can be successfully deployed in mobile apps that run efficiently with minimal computational

resources [3]. This aligns well with the goal of deploying a web-based flower recognition system for Bangladesh's local flora, providing users with an accessible platform for flower identification.

Furthermore, **“A Study on CNN-Based Transfer Learning for Recognition of Flower Species”** explores different CNN architectures, including VGG16, VGG19, DenseNet, and InceptionResNetV2. The study found that InceptionResNetV2 achieved the highest recognition accuracy of 92.25%, showcasing the benefits of advanced CNN architectures in flower species classification. The research provides valuable insights into the trade-offs between different transfer learning models and offers a solid foundation for selecting the most suitable architecture for the proposed system [4].

The research paper titled **“Flower Recognition Using CNN in Machine Learning”** discusses CNNs in the context of flower species identification, focusing on applications within the pharmaceutical and cosmetic industries. Although it does not provide specific accurate results, it underscores the growing importance of accurate plant identification systems in sectors such as medicine and cosmetics, where the correct classification of plant species is crucial for product development [5].

Moreover, the **“Flower Identification Using Machine Learning”** report from Daffodil International University introduces a real-time flower identification system for Android devices, using TensorFlow and CNNs. The app, which identifies over ten flower species with 100% accuracy for some, works offline, making it highly accessible. The report emphasizes the importance of local datasets and highlights the potential for mobile-based systems to contribute to flower recognition in Bangladesh, aligning with the objectives of this thesis to develop a similar web-based system for local flower identification [6].

By synthesizing these various approaches and leveraging insights from existing literature, this thesis aims to create a CNN-based system specifically designed for recognizing local Bangladeshi flowers. The system will utilize deep learning models, transfer learning techniques, and a user-friendly web app deployment, making flower identification accessible to a broader audience. This solution could play an important role in biodiversity conservation, environmental education, and the promotion of local plant species in Bangladesh.

This thesis will build upon the research presented in these papers by applying CNN-based approaches to the specific challenge of flower recognition in Bangladesh. By utilizing transfer learning and deploying the system as a web application, the project aims to provide an accessible, efficient, and scalable tool for local flower identification, contributing to both academic and practical applications in the field of environmental science and education.

Chapter 2

Literature Review

Deep learning, and specifically Convolutional Neural Networks (CNNs), has revolutionized the field of flower classification, enabling impressive levels of accuracy in identifying and classifying various species across diverse datasets. The ability of CNNs to learn hierarchical features from images has proven particularly effective in fine-grained flower classification tasks, where subtle variations between species need to be captured. Below are some key studies that showcase the progression and advancements in flower classification using deep learning techniques.

Rahman et al. proposed DeepFlorist, an ensemble meta-classifier that fuses features from multiple CNN models and Deep Convolutional Neural Networks (DCNNs) to enhance classification performance. By using feature fusion and classifier ensemble strategies, DeepFlorist achieved a remarkable classification accuracy of 98.4% on the Oxford Flowers102 dataset. This model's strong performance highlights the importance of combining multiple classifiers and deep features to address fine-grained flower recognition challenges [7].

Ghosh et al. introduced SA-ConvNeXt, a hybrid architecture combining the ConvNeXt model with selective attention modules. By integrating attention mechanisms that focus on the most relevant features in an image, the model was able to outperform baseline CNN architectures. This approach achieved an accuracy of 98.92% on the Oxford Flowers102 dataset, demonstrating the effectiveness of selective attention in improving the discriminative capabilities of CNNs for complex flower classification tasks [8].

Gupta et al. developed a CNN-based model specifically for classifying rose flowers. The model was tested on a specialized rose flower dataset, yielding an accuracy of 95.2%. This highlights the model's capacity for high-precision classification, especially when dealing with a single species. The focus on mono-species classification ensures that the model could be applied to domains where specific flower types are of interest, such as in botanical research and horticulture [9].

Rahimi et al. conducted a comparative study of classical CNN models, including AlexNet and VGG16, which were trained using transfer learning. The models achieved accuracies of 96.5% and 97.3%, respectively, on the Oxford Flowers102 dataset. This study validated the effectiveness of transfer learning, where pre-trained models are fine-tuned for specific tasks, significantly improving performance on flower classification tasks with relatively smaller datasets [10].

Jain et al. proposed a multiple feature fusion method using CNNs, aimed at improving flower species recognition. Their approach combined feature maps extracted at multiple layers via intersection fusion. This resulted in an impressive classification accuracy of 98.91% on the Oxford 102 dataset. Their findings demonstrate the benefit of fusing diverse feature sets to enhance the overall performance of CNN-based models, especially when working with high-dimensional datasets [11].

Kaur et al. developed a CNN-based model designed for multi-class flower classification, achieving an accuracy of 96.4%. This model has notable practical implications in fields like pharmaceuticals, cosmetics, and agriculture, where precise identification of plant species is crucial. By targeting multiple species in one model, Kaur's approach demonstrates the potential for deploying CNN models in real-world scenarios requiring high classification accuracy across several classes [12].

Hossain et al. focused on fine-tuning the VGG16 model for flower classification, achieving 97.5% accuracy on a curated dataset. Their work underscores the versatility of well-established CNN architectures, such as VGG16, in handling specialized classification tasks. By fine-tuning pre-trained models on domain-specific data, they demonstrated the model's ability to perform effectively on botanical datasets with distinct characteristics [13].

Bhuiyan et al. utilized DenseNet121, a CNN architecture known for its dense connections between layers, for flower classification. The model achieved a 97.82% accuracy on the Oxford Flowers102 dataset. DenseNet121's efficiency in terms of computational cost and model complexity, combined with high classification performance, makes it a suitable choice for applications requiring both precision and computational efficiency [14].

Begum et al. implemented a standard CNN framework for flower type recognition and achieved 95.6% accuracy. Despite being a relatively simpler approach compared to more complex architectures, this study demonstrates the continued relevance of traditional CNN architectures in capturing spatial features and patterns inherent to floral structures. This method serves as a baseline for more advanced approaches [15].

Islam et al. explored a hybrid model combining CNNs with machine learning classifiers like Support Vector Machines (SVM) and Random Forest. By optimizing hyperparameters, the model achieved 94.5% accuracy; without hyperparameter optimization, the performance dropped to 91.3%. This result underscores the importance of fine-tuning and model optimization in achieving reliable flower classification [16].

Azad et al. developed a hybrid CNN-SVM model for flower classification, which achieved 96.7% accuracy. By combining deep feature extraction from the CNN with the robustness of traditional machine learning classifiers like SVM, their approach demonstrates that hybrid models can significantly enhance classification accuracy, especially for complex tasks like flower species identification [17].

Ali et al. investigated the use of synthetic image augmentation to enhance flower classification, comparing the performance of MobileNetV2 and EfficientNetB0 models. Among the two, EfficientNetB0 achieved an accuracy of 95.6%, highlighting the potential of lightweight CNN architectures in classification tasks. This approach emphasizes the trade-off between computational efficiency and classification performance, making it suitable for deployment on mobile or edge devices [18].

Zhang et al. introduced a four-dimensional CNN model that incorporated both RGB and depth data for flower quality grading. The model achieved 93.1% accuracy, demonstrating the value of multimodal inputs in improving classification performance. This approach is particularly relevant in fields like horticultural monitoring, where additional data sources such as depth information can provide valuable insights for flower grading [19].

Singh et al. proposed SWP-LeafNET, a multistage CNN model designed for plant leaf classification, which achieved 97.4% accuracy. While primarily focused on leaves, the model's techniques and insights are transferable to flower classification tasks, providing valuable approaches for handling different types of botanical classification problems [20].

Nguyen et al. presented a CNN-based system for identifying Vietnamese flora, achieving 94.6% accuracy. Their work emphasizes the adaptability of deep learning models to region-specific flora, making it an essential tool for local biodiversity studies and conservation efforts [21].

Li et al. used Faster R-CNN with the Inception-ResNet v2 architecture to monitor cotton flowering, achieving 92.8% detection accuracy. This study highlights the application of deep learning in precision agriculture, where accurate detection of flowering events is crucial for optimizing crop yields and ensuring agricultural sustainability [22].

Sharma et al. conducted a comparative study of several CNN architectures for flower species classification. They found that ResNet50 performed the best, achieving 96.9% accuracy on the Oxford Flowers102 dataset. This study is significant as it provides a deeper understanding of which architectures are best suited for specific flower classification tasks [23].

Banu et al. proposed a deep learning model specifically tailored for floral classification, which achieved an accuracy of 97.1%. Their work underscores the importance of customizing CNN architectures to optimize them for specific domain datasets, ensuring higher accuracy in specialized flower identification tasks [24].

Basnet et al. employed a conventional CNN framework for flower species classification in Nepal, reporting an accuracy of 93.8%. Their work contributes to efforts focused on local biodiversity identification, offering a practical solution for regions with limited resources [25].

We can conclude that no significant research has been conducted on the classification of local-colored flowers in Bangladesh, especially with respect to the unique visual characteristics and regional diversity of floral species. Most existing studies have concentrated on global flower datasets such as Oxford Flowers102, which do not represent the indigenous flora of Bangladesh. Furthermore, only a very limited number of research efforts have explored flower classification using locally collected data in this region. Our study aims to address this gap by focusing on local-colored flowers found in Bangladeshi markets and gardens, using a regionally curated dataset to develop and evaluate deep learning models for accurate classification.

Table 2.1: Tabular list of the related works.

#	Paper Title	Model Used	Accuracy	Advantages	Disadvantages
1	Flower Species Identification Using Deep Convolutional Neural Networks [1]	Comparative study of AlexNet, CaffeNet, and GoogLeNet	GoogLeNet achieved 67.45% at Rank-1 and 90.82% at Rank-10 on the PlantCLEF 2015 dataset	Outperformed traditional methods like Kernel Descriptor (KDES) by a significant margin, No need for handcrafted features or preprocessing	Moderate accuracy at Rank-1, indicating challenges in fine-grained classification across 967 species, Performance may vary with different datasets and conditions

2	Flower Species Recognition System Using Convolution Neural Networks and Transfer Learning [2]	Transfer learning with OverFeat, Inception-v3, and Xception	Inception-v3: 93.41% Xception: 90.60% OverFeat: 73.05% on the FLOWER S 102 datasets	Transfer learning reduces the need for extensive training data and computational resources, High accuracy achieved with pre-trained models	Dependence on pre-trained models may limit adaptability to new or significantly different datasets, Performance is influenced by the choice of classifier used on top of the feature extractor
3	Recognition of Flowers Using Convolutional Neural Networks [3]	Convolutional Neural Networks (specific architecture not specified)	83.13%	Mobile application development; practical implementation	Limited to four flower categories, which may not generalize well to other species, Specific model architecture and training details are not provided, limiting reproducibility
4	A Study on CNN Based Transfer Learning for	VGG16, VGG19, DenseNet, and InceptionResNet V2	92.25%,	Explores transfer learning effectiveness	Insufficient details on model and performance

	Recognition of Flower Species [4]				
5	Flower Recognition Using CNN in Machine Learning [5]	CNN (architecture unspecified)	Not specified	Application of CNNs in flower recognition	Lacks specific model and performance information
6	Flower Identification Using Machine Learning [6]	CNN implemented with TensorFlow	Up to 100% for some flowers; varies overall	Real-time identification; focused on Bangladeshi flowers	Inconsistent accuracy; limited flower categories
7	DeepFlorist: Deep Convolutional Ensemble for FineGrained Flower Classification [7]	Ensemble of CNN + DCNN (with feature fusion)	98.4%	High accuracy in fine-grained classification, robust generalization	High computational cost due to ensemble approach
8	SA-ConvNeXt: A Selective Attention ConvNeXt Hybrid for Flower Classification	ConvNeXt + Selective Attention	98.92%	Superior feature discrimination, high accuracy	Increased model complexity, harder to interpret

	n [8]				
9	Rose Classification Using CNN [9]	CNN	95.2%	Accurate mono-species classification, fast training	Limited to roses, poor generalizability
10	Comparative Study of CNN Models for Flower Classification [10]	AlexNet, VGG16	96.5% (AlexNet), 97.3% (VGG16)	Validates pre-trained models, effective transfer learning	Dependent on ImageNet features
11	Multiple Feature Fusion for Flower Species Recognition [11]	CNN + Intersection Fusion of Feature Maps	98.91%	Combines feature richness, high accuracy	Complex training pipeline
12	Multi-Class Flower Classification using CNN [12]	CNN	96.4%	Efficient for multi-class tasks, practical usability	Moderate accuracy, not state-of-the-art
13	Fine-Tuned VGG16 for Flower Classification [13]	VGG16 (Fine-tuned)	97.5%	Effective on smaller datasets, simple to deploy	Limited scalability
14	DenseNet121 for Flower Classification	DenseNet121	97.82%	High accuracy, reduced overfitting	Dense connections increase memory

	n [14]				usage
15	CNN Framework for Flower Type Recognition [15]	Standard CNN	95.6%	Straightforward to implement, good baseline	Underperforms on fine-grained species
16	Hybrid CNN-SVM for Flower Classification [16]	CNN + SVM / Random Forest	94.5% (with tuning), 91.3% (without)	Fusion of deep and classical methods, robust	Requires hyperparameter tuning, complexity rises
17	Hybrid CNN-SVM Model [17]	CNN + SVM	96.7%	Balanced feature learning and classification	May not generalize across domains
18	Comparison of MobileNetV2 and EfficientNet B0 [18]	MobileNetV2, EfficientNetB0	95.6% (Efficient Net)	Lightweight, fast inference	Slight drop in accuracy vs. heavier models
19	4D CNN for Flower Quality Grading [19]	4D CNN (RGB + Depth)	93.1%	Integrates spatial-depth cues, robust grading	Requires specialized sensors for data
20	SWP-LeafNET [20]	Multistage CNN	97.4%	Layered architecture enables fine detail capture	Designed for leaves, not flowers

21	CNN for Vietnamese Flora Classification [21]	CNN	94.6%	Customized for local flora, practical use	Not easily generalizable
22	Faster R-CNN with Inception-ResNet v2 for Cotton Flower Monitoring [22]	Faster R-CNN + Inception-ResNet v2	92.8%	High precision for detection	High computation and resource requirement
22	CNNs for Flower Species and Chemical Profiling [23]	ResNet50	96.9%	Combines visual and chemical data, dual capability	Complex integration process
24	Optimized CNN for Floral Classification [24]	Custom Optimized CNN	97.1%	Built specifically for flowers, high accuracy	Lacks adaptability to other domains
25	Implementation and assessment of new hybrid model using CNN for flower image classification [25]	CNN	93.8%	Supports local biodiversity, low resource	Poor cross-domain performance

Chapter 3

Research Questions and Objectives

In this study, we aim to apply a Convolutional Neural Network (CNN) model to classify images of locally grown colored flowers in Bangladesh into distinct categories based on their species and visual characteristics, leveraging deep learning techniques.

3.1 Research Questions:

The questions we are considering for this study are as follows:

1. What methods can be applied to accurately classify colored flower species found in Bangladesh?

Classifying Bangladeshi flowers involves a series of steps, including image collection from regional sources, preprocessing, feature extraction, model selection (such as Convolutional Neural Networks), training and evaluation, and deployment through a user-friendly web application for real-time usage.

2. How can local-colored flowers be classified autonomously using deep learning techniques?

Autonomous flower classification can be achieved through a pipeline involving image acquisition, preprocessing, data augmentation, CNN model training (e.g., using architectures like MobileNet, EfficientNet, or Inception etc.), and integration into a responsive web interface for instant classification and species recognition.

3. What image processing steps are essential before feeding local flower images into deep learning models?

To optimize the performance of CNNs, key preprocessing techniques include image resizing, normalization, background filtering, contrast enhancement, color correction, noise removal, and augmentation strategies such as flipping, rotation, and scaling. These steps ensure robust feature extraction for improved classification accuracy.

4. Is there a dedicated, high-quality dataset available for classifying colored flowers local to Bangladesh?

Yes, the “**ColoredFlowersBD**” dataset is a curated collection of images featuring 13 different types of flowers found in Bangladesh. This dataset was collected and organized by our team with the goal of supporting research in floral image analysis. Each class in the dataset represents a

specific flower type, captured in both single and bulk forms. It comprises high-resolution, annotated images of local flowers from various regions in Bangladesh, capturing their color diversity, natural lighting conditions, and seasonal variations to support training of deep learning models tailored for local classification tasks.

5. How can automated flower classification be integrated into a web-based platform for public or institutional use?

Automating flower recognition through a CNN-powered web application offers scalable and real-time access to species identification for educational, agricultural, and botanical research use cases. The deployment enables user-friendly interaction, high-speed image inference, and backend integration with trained CNN models for seamless predictions.

6. What are the potential industrial and commercial applications of automated local flower classification?

Automated flower classification can significantly benefit industries such as agriculture, floriculture, cosmetics, herbal medicine, and landscaping. It allows for efficient flower grading, inventory management, quality control in export processes, automated cataloging in nurseries, and visual search in online floral marketplaces. Integration with IoT devices and smart agriculture systems could further enhance crop monitoring and flower health assessment.

7. How can this system contribute to biodiversity documentation and educational awareness in Bangladesh?

The integration of flower recognition into a digital platform helps preserve botanical knowledge, aids researchers in field identification, and enhances public education about native flora. The model can also support environmental monitoring, pollination tracking, and urban landscaping efforts through easy-to-use visual recognition tools.

3.2 Research Objectives:

1. To develop an efficient CNN-based deep learning model capable of accurately classifying and recognizing various species of colored flowers local to Bangladesh using the ColoredFlowersBD dataset.

2. To analyze and identify the most effective CNN architectures (such as MobileNet, EfficientNet, ResNet, etc.) in terms of accuracy, speed, and computational efficiency for the flower classification task.
3. To perform extensive image preprocessing and augmentation techniques to enhance model generalization and robustness against variations in lighting, orientation, and background in the collected flower images.
4. To evaluate the performance of the model using appropriate metrics such as accuracy, precision, recall, and F1-score across different flower species in the dataset.
5. To deploy the trained model as a user-friendly web application, allowing users (e.g., researchers, hobbyists, agricultural professionals) to upload flower images and receive instant classification results.
6. To explore the real-world applicability and industrial relevance of automated flower recognition systems in fields such as horticulture, agriculture, floriculture, botanical research, and e-commerce platforms for local flower trading.

Chapter 4

Materials & Proposed Methodology

4.1 Materials

Colored Flower Recognition is a collection of images of 13 different types of local colored flower. We take images of flowers using smartphones in different weather. The collection of 7993 raw images was prepared and specific features were extracted by hand. The method uses five different pre-trained CNN models to accurately identify plants. The models are MobilenetV2, ResNet50V2, EfficientNetv2B0, InceptionV3 and Xception. The easy-to-use “**CNN-Based Computer Vision Approach for Local Bangladeshi Colored Flower Recognition**” helps identify flowers in real time and supports plant classification research.

The workflow of the data collection process is shown in Figure 3.1. Firstly, we figured out a specific location. We made sure that we have a suitable camera to capture high-quality images, and lastly, we pre-processed our images for the project.

4.1.1 Dataset Collection

The “**ColoredFlowersBD**” dataset is a curated collection of images featuring 13 different types of flowers found in Bangladesh. This dataset was collected and organized by our team with the goal of supporting research in floral image analysis. Each class in the dataset represents a specific flower type, captured in both single and bulk forms, offering a rich variety in terms of color, shape, and appearance. To detect and classify colored flowers accurately, high-quality image samples were essential. We collected samples from various locations, including the Botanical Garden, Bangladesh Agricultural University, Narsingdi, and several local nurseries across Dhaka city. These diverse locations were chosen to capture a wide range of flower types, colors, and natural variations.

To ensure high-quality and consistent data, we used an iPhone 15 smartphone capable of capturing high-resolution images. Each flower was photographed from nine different angles to reduce the effects of lighting variations such as direct sunlight or shadows. Images were taken in a controlled laboratory environment to preserve the natural color of the flowers, which was crucial for reliable classification.

The image collection process followed these steps:

1. Discuss with nursery staff to understand flower types and availability.
2. Observe the condition and variety of different colored flowers.
3. Select appropriate locations with optimal natural lighting.
4. Capture images from nine different angles for each flower.
5. Preprocess the data by resizing images and ensuring relevance to colored flower classification.

To ensure accessibility and reproducibility of this work, the “ColoredFlowersBD” dataset has been made publicly available on **Mendeley Data**. It can be accessed via the following link: [“https://data.mendeley.com/datasets/d22kghy9xb”](https://data.mendeley.com/datasets/d22kghy9xb)

A screenshot of the dataset’s Mendeley Data page is provided below for reference:



ColoredFlowersBD: A Comprehensive Image Dataset of Colored Flowers in Bangladesh for Identification and Classification Using Machine Learning and Computer Vision

Published: 19 February 2025 | Version 1 | DOI: 10.17632/d22kghy9xb.1

Contributors: Md Hasanul Ferdaus, Rizvee Hassan Prito, Masud Ahmed, Tuly Rahman, Riya Saha, Kazi Minhazul Goni Sami, Md Sajjad Hossain

Description

Type of data: 720 x 720 px images of colored flowers.

Data format: JPEG

Dataset contents: Original images of different varieties of colored flowers in Bangladesh from single flower and bulk-flower perspectives.

Number of classes: Thirteen colored flower varieties - (1) Chandramallika, (2) Cosmos Phul, (3) Gada, (4) Golap, (5) Jaba, (6) Kagoj Phul, (7) Noyontara, (8) Radhachura, (9) Rangan, (10) Salvia, (11) Sandhyamani, (12) Surjomukhi, and (13) Zinnia.

Figure 4.1: Screenshot of the “ColoredFlowersBD” dataset on the Mendeley Data platform.

Here is the summary of the full data collection process:

Table 4.1: Data collection summary

Data Attributes		Amount and Specification	
Name of Flowers		Single	Bulk
1. Chandramallika		306	314
2. Cosmos Phul		307	313
3. Gada		304	313
4. Golap		302	303
5. Jaba		300	304
6. Kagoj Phul		301	311
7. Noyontara		303	306
8. Radhachura		309	308
9. Rangan		305	301
10. Salvia		313	321
11. Sandhyamani		305	310
12. Surjomukhi		310	311
13. Zinnia		307	306
Total		7993	
Time		Daylight	
Captured Objects' View		From 9 angles	
Places		Botanical Garden, Bangladesh Agricultural University, Narsingdi, Uttara Dia Bari and several local nurseries across Dhaka city	
Image quality		Dimension: 720x 720 pixels Format: JPG Quality: standard	
Device used		 iPhone 15 - 48 MP, f/1.6, 26mm (wide), 1/1.56", 1.0µm, dual pixel PDAF, sensor-shift OIS  Poco X3 - 64 MP, f/1.9, (wide), 1/1.73", 0.8µm, PDAF	

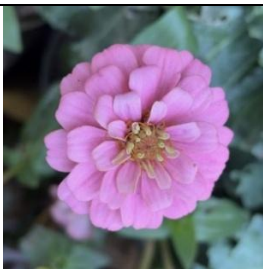
4.1.2 Dataset Sampling:

Here is the sample figure of those flowers is given Below: The dataset contains images of Thirteen Classes where Single and Bulk 2 Classes are Available.

Table 4.2: Data Sampling

Name of Flowers	Scientific Name	Common English Name	Single Flower	Bulk Flower
1.Chandrama llika	1.Chrysanthemum indicum	1.Chrysanthemum		
2.Cosmos Phul	2.Cosmos bipinnatus	2. Cosmos		
3.Gada	3. Tagetes erecta	3. Marigold		
4.Golap	4. Rosa spp.	4. Rose		

5.Jaba	5. Hibiscus rosa-sinensis	5. China rose		
6.Kagoj Phul	6. Bougainvillea glabra	6. Bougainvillea		
7.Noyontara	7.Catharanthus roseus	7. Periwinkle		
8.Radhachura	8.Caesalpinia pulcherrima	8. Peacock Flower		
9.Rangan	9. Ixora coccinea	9. Jungle Geranium		
10.Salvia	10.Salvia splendens	10. Sage		

11.Sandhyamani	11. Ruellia tuberosa	11. Four O'Clock Flower		
12.Surjomukhi	12. Helianthus annuus	12. Sunflower		
13.Zinnia	13. Zinnia elegans	13. Zinnia		

4.1.3 Dataset Exploration

A large collection of high-quality photos has been assembled to aid in the creation of sophisticated image analysis software. More than [7993] photos of Colored flowers are included in the dataset.

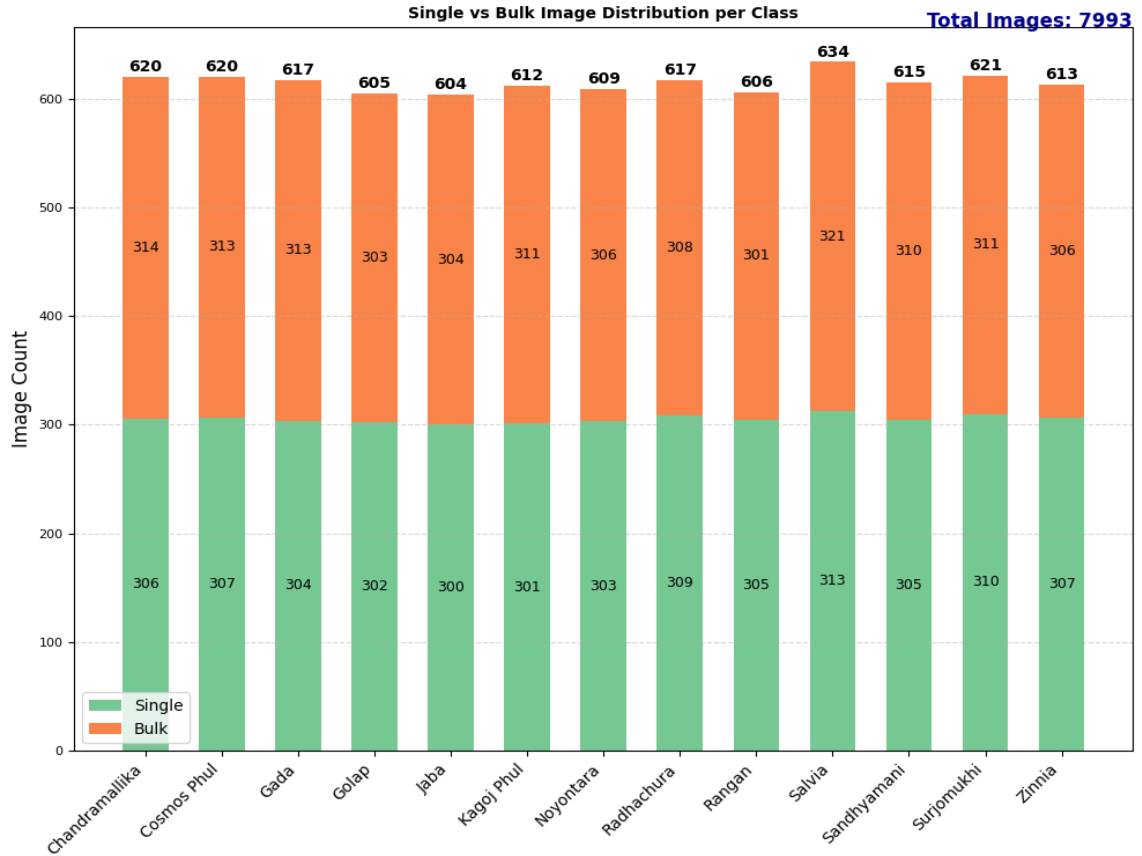


Figure 4.2: Structure of the Bangladeshi Colored Flowers Dataset

The data distribution of each class is almost the same. There was no need to employ any additional strategies to address the data imbalance because there were no serious issues with it in the dataset. No missing value has been found.

4.1.4 Dataset Pre-Processing

To enhance the performance and generalization capability of our deep learning model, we applied a series of preprocessing and augmentation techniques to the image dataset using the Roboflow platform. These steps are essential for improving the robustness of the model, particularly when deployed on unseen or real-world data. Data augmentation increases the variability of the training set without collecting new data, effectively reducing overfitting and enabling better generalization. Table 4.3 presents the augmentation methods applied to the dataset. Each method was carefully selected to simulate different real-world conditions and distortions, making the model more resilient to common variations such as lighting changes, orientation shifts, and noise.

Table 4.3: Augmentation Methods

Methods	Values
Flip	Horizontal, Vertical
Rotation	Between -15 degree and +15 degree
90 Degree Rotate	Clockwise, Counter-Clockwise, Upside Down
Brightness	Between -20% and +20%
Noise	Up to 2% of pixels
Blur	Up to 3px
Image After Augmentation	20728

Description of Techniques:

- **Flip:** Horizontal and vertical flipping introduces orientation variations, helping the model generalize across differently aligned images.
- **Rotation:** Small angle rotations between -15° and $+15^\circ$ simulate slight camera tilts or variations in image capture angles.
- **90 Degree Rotation:** Clockwise, counter-clockwise, and upside-down rotations allow the model to handle extreme orientation changes.
- **Brightness Adjustment:** Altering brightness mimics varying lighting conditions, such as shadows or overexposure.
- **Noise Injection:** Random noise simulates real-world image corruption, training the model to focus on key features rather than irrelevant artifacts.
- **Blur:** Applying up to 3-pixel blur simulates out-of-focus images or motion blur, ensuring model performance on imperfect images.

After applying these transformations, the dataset size increased to a total of **20728** images, significantly enriching the diversity and variability of training data.

4.2 Model Selection and Training

Our research journey started with the collection of high-quality colored flower images. We were looking for a suitable garden field for our project and we discussed different kinds of flowers with gardeners. We carefully separated single and bulk classes. We captured each flower from 9 different angles to get the best view of the disease. Our dataset is well distributed. We leveled our images so that it will be easier to find out from the original folder. For image preprocessing we applied resize so that all images will be in same sizes. Then we applied brightness adjustments. Following preprocessing, our approach to detect leaf disease using deep learning techniques. For the deep learning segment, leaf images were transformed using techniques like normalization and segmentation. To augment dataset size and introduce variability, we employed image augmentation techniques such as flipping, zooming, and shearing. Post-augmentation, we applied pre-trained models like, MobileNetV2, EfficientNetV2B0, ResNet50V2, InceptionV3 and Xception.

The performance of the model was evaluated using key metrics including Accuracy, Precision, Recall, and F1-Score. These metrics provide a comprehensive view of the model's effectiveness, especially in handling class imbalances and capturing both the correctness and completeness of predictions. The dataset was split into three parts: 80% for training, 10% for validation, and 10% for testing. This stratified division ensured a balanced representation of data in each subset, enabling the model to generalize effectively while being validated and tested on unseen samples.

For optimization, the Adam optimizer was employed due to its adaptive learning rate and efficiency in handling sparse gradients. The categorical cross-entropy loss function was used, which is well-suited for multi-class classification problems. The model was trained over multiple epochs, with early stopping implemented to monitor validation loss and halt training when performance began to degrade, thus preventing overfitting.

4.2.1 Design and Implementation

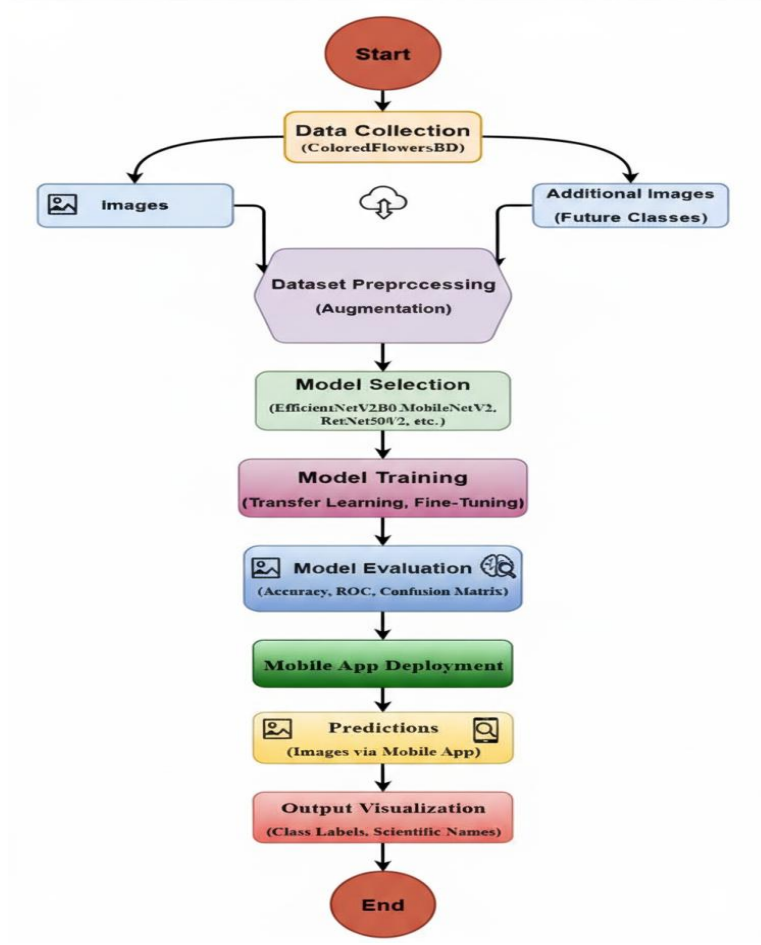


Figure 4.3: Flowchart of the whole process.

4.2.2 Proposed Model

The proposed model for local Bangladeshi Colored Flower recognition leverages advanced deep learning techniques to accurately classify and identify flowers images. The methodology begins with preprocessing, where flower images are normalized and segmented to enhance their quality and focus on the relevant features.

To address the variability and enhance the robustness of the model, we employ various image augmentation techniques. These techniques include flipping, zooming, and shearing, which effectively increase the dataset size and introduce diverse perspectives of the colored flowers images, thereby improving the model's generalization capabilities.

1. MobileNetV2:

MobileNetV2 is a lightweight and efficient convolutional neural network architecture specifically designed for mobile and embedded vision applications. It employs depth wise separable convolutions to significantly reduce computational cost. A key innovation in MobileNetV2 is the use of inverted residual blocks with linear bottlenecks. Each block consists of a 1×1 convolution for channel expansion, a depth wise convolution for spatial filtering, and another 1×1 convolution for projection to a lower-dimensional space. Residual connections are used when the input and output dimensions match, enhancing gradient flow while keeping the model compact and efficient.

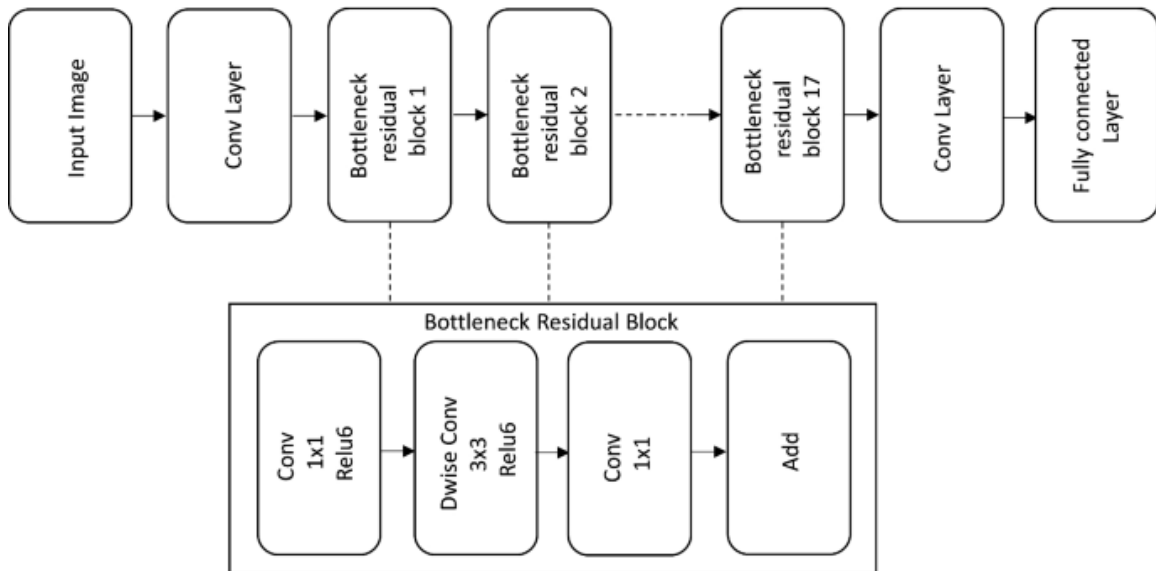


Figure 4.4: Architecture of MobileNetV2

2. ResNet50V2:

ResNet50V2 is an enhanced version of the original ResNet50, known for its introduction of residual connections that mitigate the vanishing gradient problem in deep networks. ResNet50V2 adopts a pre-activation layout, where batch normalization and ReLU activation are applied before each convolution, rather than after. The network is constructed using bottleneck blocks, each composed of three convolutional layers (1×1 , 3×3 , and 1×1), with identity or projection shortcuts enabling deeper networks to train more effectively and efficiently.

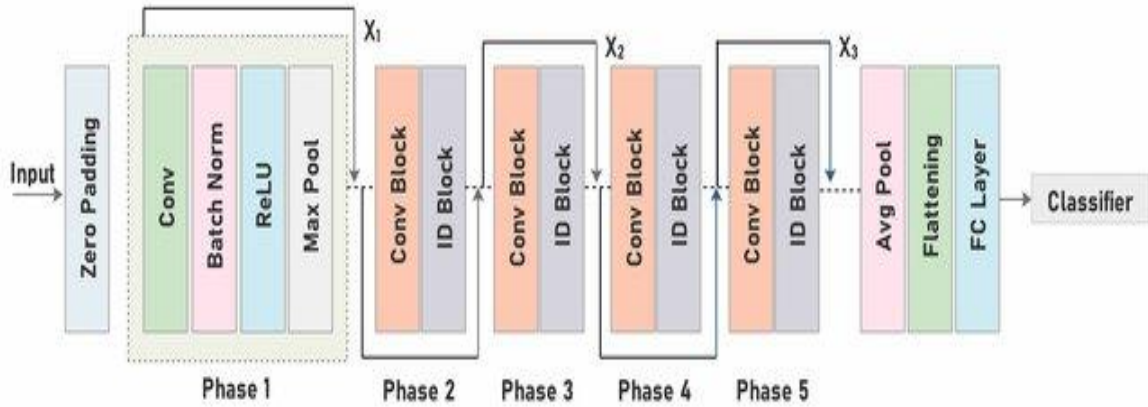


Figure 4.5: Architecture of ResNet50V2

3. EfficientNetV2B0:

EfficientNetV2B0 is a modern and highly efficient convolutional neural network that improves upon the original Efficient Net series by optimizing both training speed and model accuracy. It utilizes a mix of Fused-MBConv and MBConv blocks—both variations of inverted residual blocks—along with squeeze-and-excitation (SE) attention mechanisms. The architecture applies compound scaling to systematically balance network depth, width, and input resolution. EfficientNetV2B0 also adopts SiLU activations and batch normalization throughout, making it well-suited for both resource-constrained environments and high-performance tasks.



Figure 4.6: Architecture of EfficientNetV2B0

4. InceptionV3:

InceptionV3 builds on the earlier Inception models by introducing several innovations such as *factorized convolutions*, *auxiliary classifiers*, and *efficient grid size reduction*. The core component of InceptionV3 is the *Inception module*, which processes inputs through multiple parallel convolution paths (e.g., 1×1 , 3×3 , 5×5 convolutions and pooling) and concatenates the outputs. Factorizing convolutions (e.g., 3×3 into two $1D$ convolutions) reduces computational costs. Auxiliary classifiers act as regularizers and help with gradient flow during training. Overall, InceptionV3 strikes a strong balance between accuracy and computational efficiency.

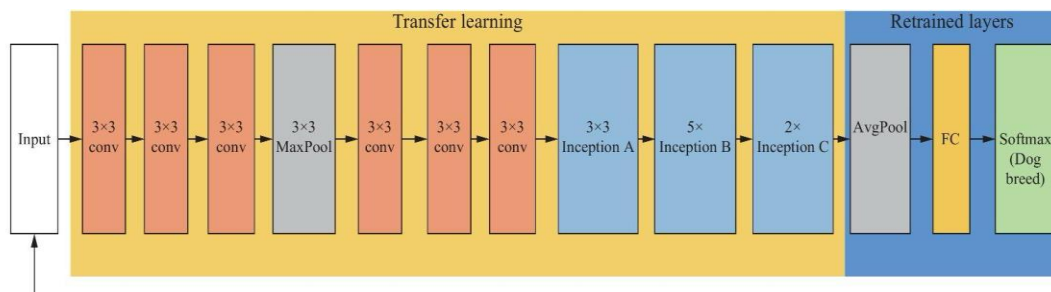


Figure 4.7: Architecture of InceptionV3

5. Xception:

Xception (Extreme Inception) is a deep convolutional neural network that extends the Inception architecture by fully embracing *depth wise separable convolutions*. Unlike Inception modules that mix different convolution types, Xception replaces them with a series of separable convolutions combined with *residual connections*. The architecture is divided into three main flows: entry, middle, and exit. The middle flow comprises repeated identical modules, while residual connections maintain gradient stability across deep layers. Xception achieves high accuracy while maintaining computational efficiency, often outperforming InceptionV3 in benchmark tasks.

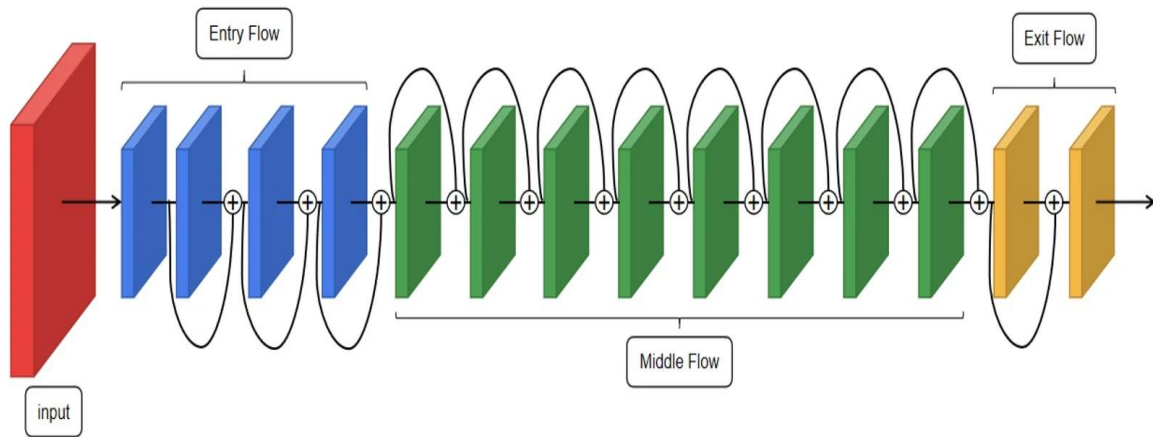


Figure 4.8: Architecture of Xception

4.3 Implementation Environment

We have worked on a total of five deep learning models which were implemented in most of the previous research lately for detecting different image types precisely. We trained the models in the Google Colab environment, a running environment for running python codes and our training was done on the Tesla T4 GPU card, with the GPU memory size of 15 GB, 12.7 GB of system memory, and the disk size of 78.2 GB. Methods of investigated prediction in the testing phase, which are often equivalent to training for more valuation.

Chapter 5

Results and Discussion

5.1 Results obtained

This section compares and displays the outcomes that our models produced. We used 5 Pre-trained CNN models MobileNetV2, EfficientNetV2B0, ResNet50V2, InceptionV3 and Xception. We assessed our results using the testing metrics of accuracy, precision, recall, and f1-score. We can see that models have performed quite well. The testing accuracy of EfficientNetV2B0 is the highest which is 99.75% for both single flowers and bulk flowers and the lowest is InceptionV3 for Single flowers which is 95.71% and InceptionV3 for bulk flowers which is 86.07%.

The following table and Figure 4.1 show the obtained results for the machine learning models of single flowers.

Table 5.1: Model Accuracy, Efficiency, and Size for Single Flowers Classification

Model	Model Size	Accuracy	Loss	Precision	Re-call	F1-score
MobileNetV2	8.68 MB	99.24%	0.0400	0.99	0.99	0.99
EfficientNetV2B0	22.64 MB	99.75%	0.0061	1.0	1.0	1.0
ResNet50V2	89.99 MB	97.73%	0.0862	0.98	0.98	0.98
InceptionV3	83.27 MB	95.71%	0.1446	0.96	0.96	0.95
Xception	79.68 MB	99.49%	0.0223	1.0	0.99	1.0

Model Comparison:

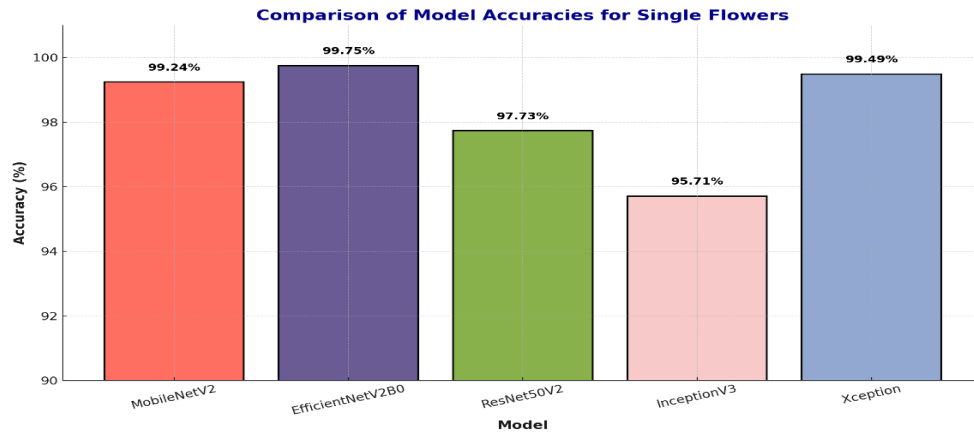


Figure 5.1: Comparison of Model Accuracies for Single Flowers

From all the models of Single flowers, we can see that EfficientNetV2B0 has the highest accuracy, precision, recall, and f1-score.

Here is another following table and Figure 4.2 show the obtained results for the machine learning models of bulk flowers.

Table 5.2: Model Accuracy, Efficiency, and Size for Bulk Flowers Classification

Model	Model Size	Accuracy	Loss	Precision	Re-call	F1-score
MobileNetV2	8.68 MB	98.01%	0.0584	0.98	0.98	0.98
EfficientNetV2B0	22.64 MB	99.75%	0.0041	1.0	1.0	1.0
ResNet50V2	89.99 MB	95.52%	0.1414	0.95	0.95	0.95
InceptionV3	83.27 MB	86.07%	3.9409	0.88	0.86	0.86
Xception	79.68 MB	99.51%	0.0799	0.98	0.99	0.99

Model Comparison:

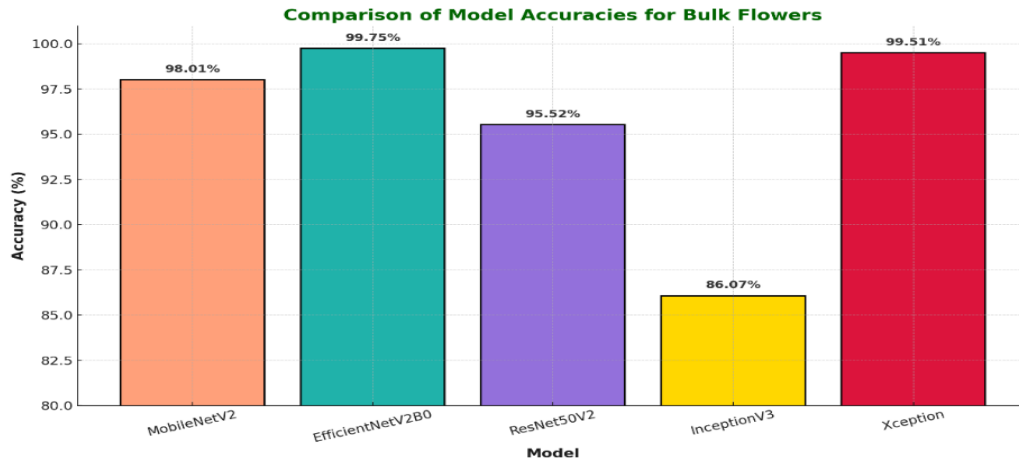


Figure 5.2: Comparison of Model Accuracies for Bulk Flowers

Of all the models of Bulk flowers, we can see that EfficientNetV2B0 has the highest accuracy, precision, recall, and f1-score.

5.2 In-Depth Result Analysis

To evaluate the performance of the proposed model, we conducted a thorough analysis using key metrics and visualization tools. These include the training vs. validation loss curve, ROC curve, and the confusion matrix.

5.2.1. Training vs Validation Loss Curve: The training and validation loss curve provides insights into how well the model is learning and generalizing to unseen data. An ideal curve shows a decreasing trend for both losses, with validation loss stabilizing close to the training loss. A large gap or an increase in validation loss while training loss continues to decrease may indicate overfitting.

❖ Training vs Validation Loss Curve for Single Flowers:

1. MobileNetV2:

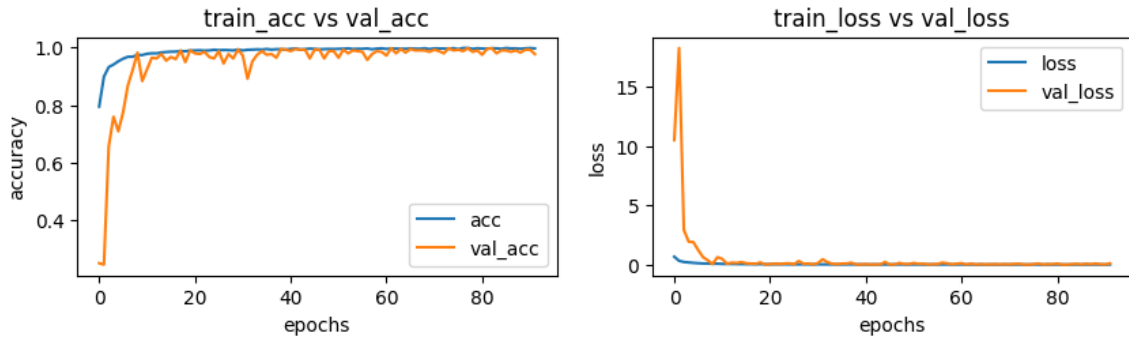


Figure 5.3: Training Accuracy and Loss Curve of MobileNetV2 for Single Flowers

2. EfficientNetV2B0:

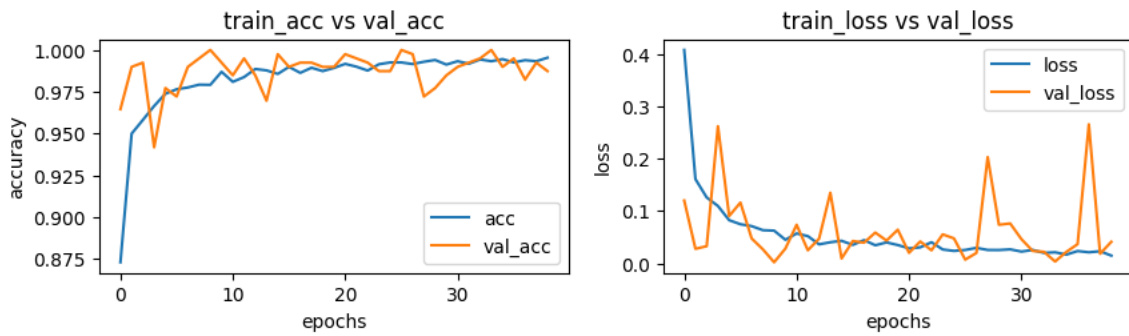


Figure 5.4: Training Accuracy and Loss Curve of EfficientNetV2B0 for Single Flowers

3. ResNet50V2:

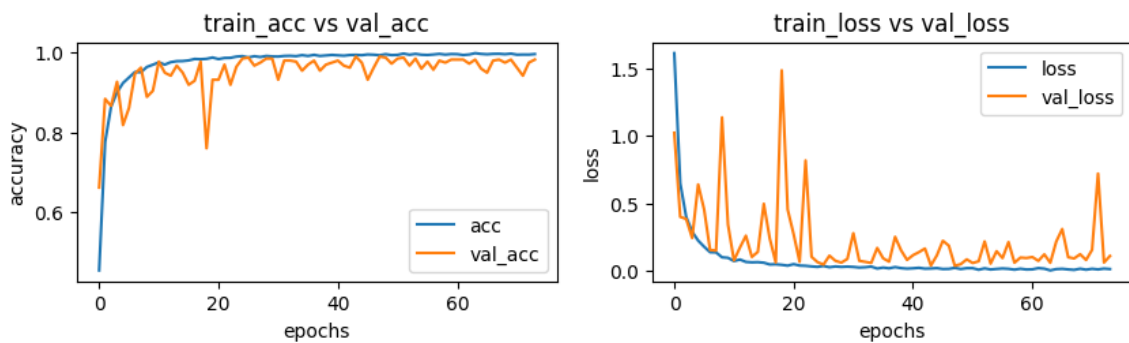


Figure 5.5: Training Accuracy and Loss Curve of ResNet50V2 for Single Flowers

4.InceptionV3:

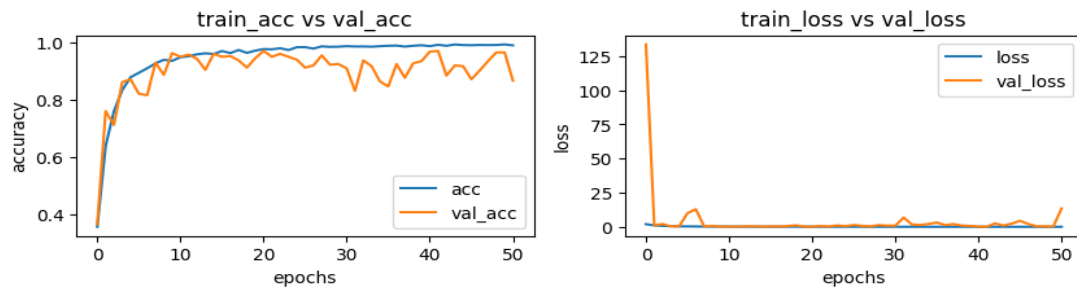


Figure 5.6: Training Accuracy and Loss Curve of InceptionV3 for Single Flowers

5.Xception:

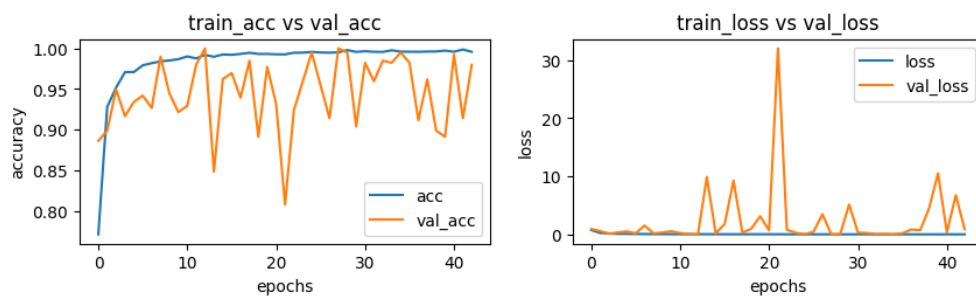


Figure 5.7: Training Accuracy and Loss Curve of Xception for Single Flower

❖ Training vs Validation Loss Curve for Bulk Flowers:

1. MobileNetV2:

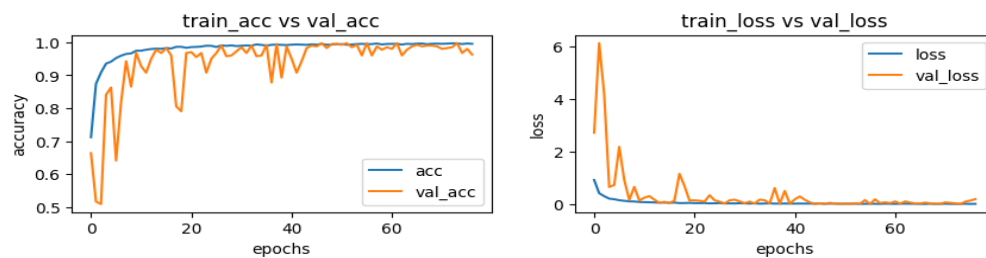


Figure 5.8: Training Accuracy and Loss Curve of MobileNetV2 for Bulk Flowers

2. EfficientNetV2B0:

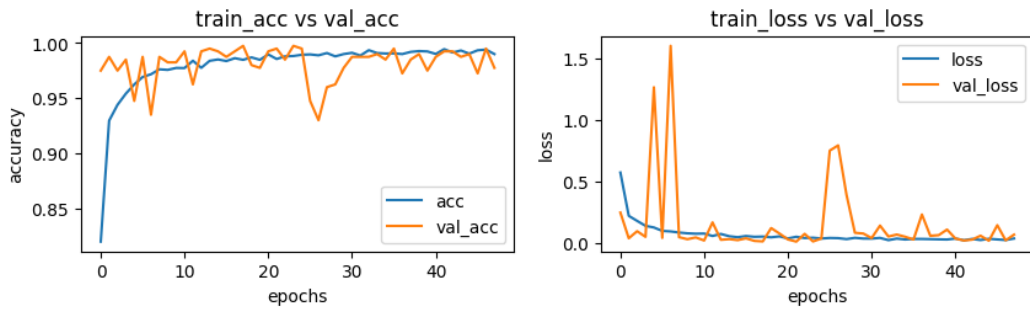


Figure 5.9: Training Accuracy and Loss Curve of EfficientNetV2B0 for Bulk Flowers

3. ResNet50V2:

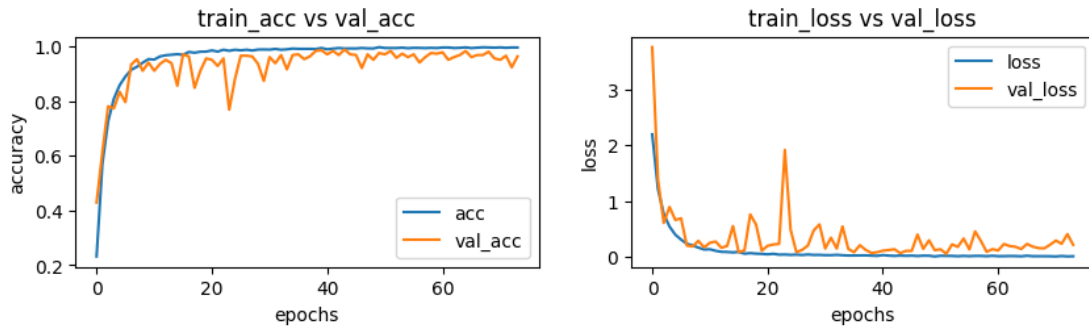


Figure 5.10: Training Accuracy and Loss Curve of ResNet50V2 for Bulk Flowers

4. InceptionV3:

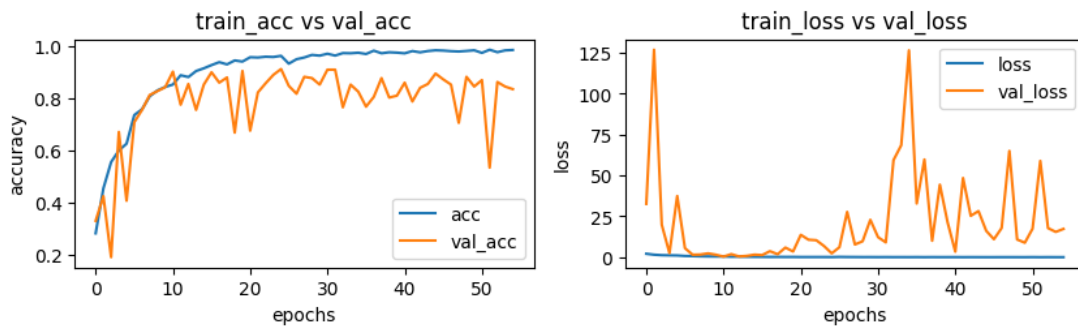


Figure 5.11: Training Accuracy and Loss Curve of InceptionV3 for Bulk Flowers

5.Xception:

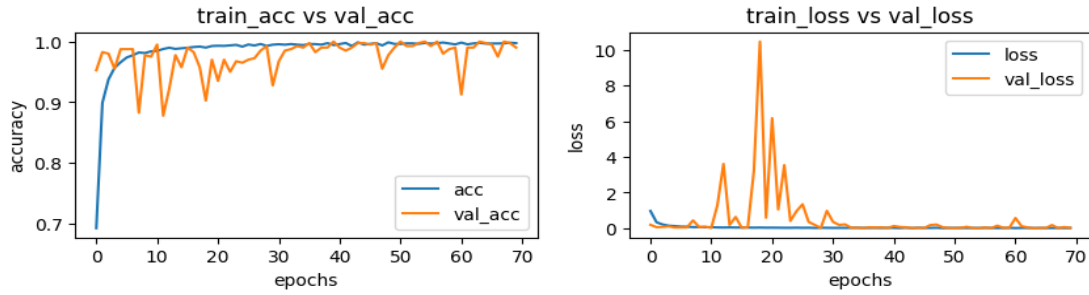


Figure 5.12: Training Accuracy and Loss Curve of Xception for Bulk Flowers

Observations:

- The training loss decreases steadily, indicating effective learning.
- The validation loss follows a similar trend with minor fluctuations, suggesting good generalization performance.
- No significant overfitting is observed.

5.2.2 ROC Curve

The Receiver Operating Characteristic (ROC) curve evaluates the trade-off between the true positive rate (sensitivity) and the false positive rate across different classification thresholds. The area under the ROC curve (AUC) is a measure of the model's ability to distinguish between the classes.

❖ ROC curve for Single flowers:

1. MobileNetV2:

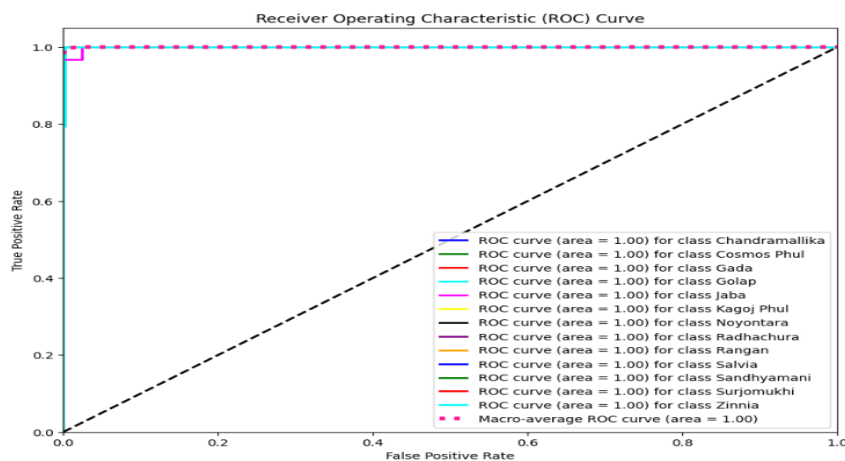


Figure 5.13: ROC Curve of MobileNetV2 for Single Flowers

2. EfficientNetV2B0:

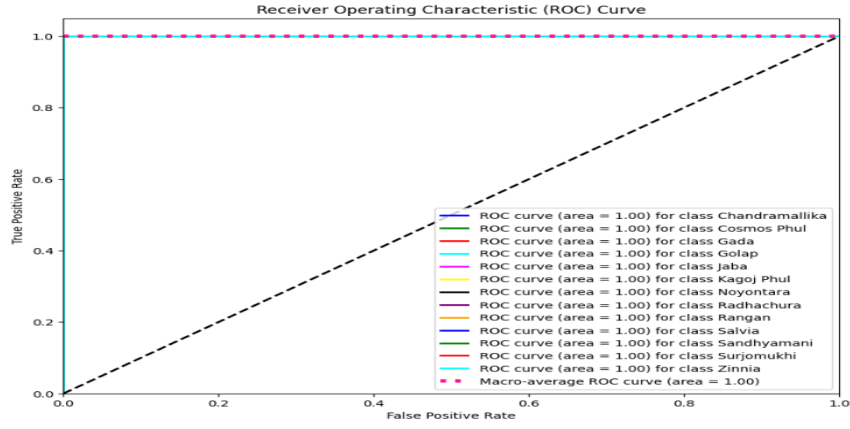


Figure 5.14: ROC Curve of EfficientNetV2B0 for Single Flowers

3. ResNet50V2:

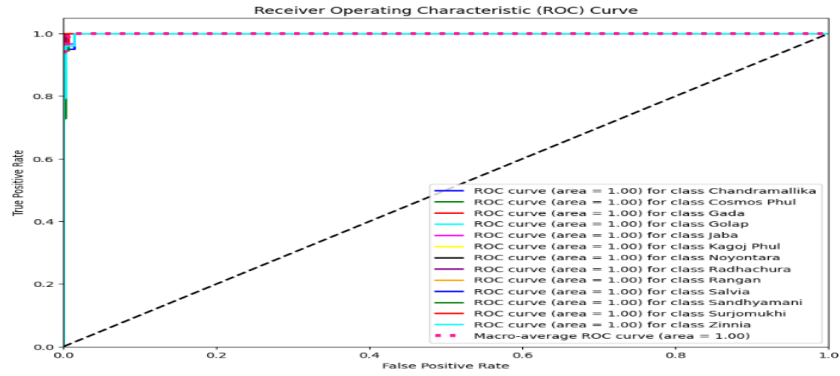


Figure 5.15: ROC Curve of ResNet50V2 for Single Flowers

4. InceptionV3:

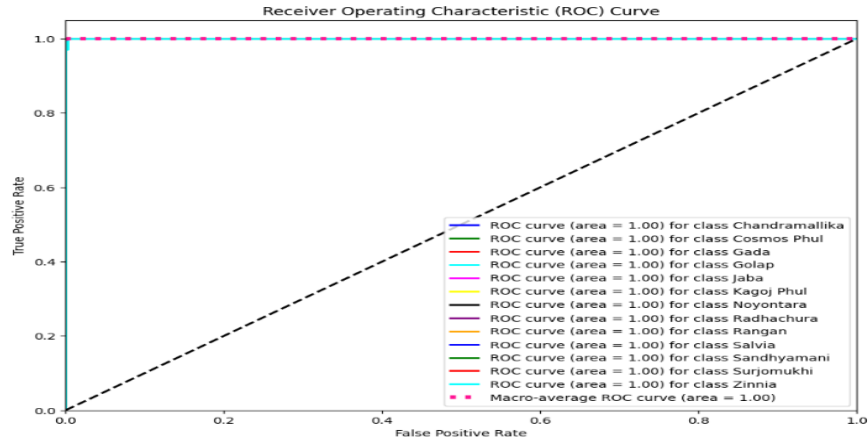


Figure 5.16: ROC Curve of InceptionV3 for Single Flowers

5. Xception:

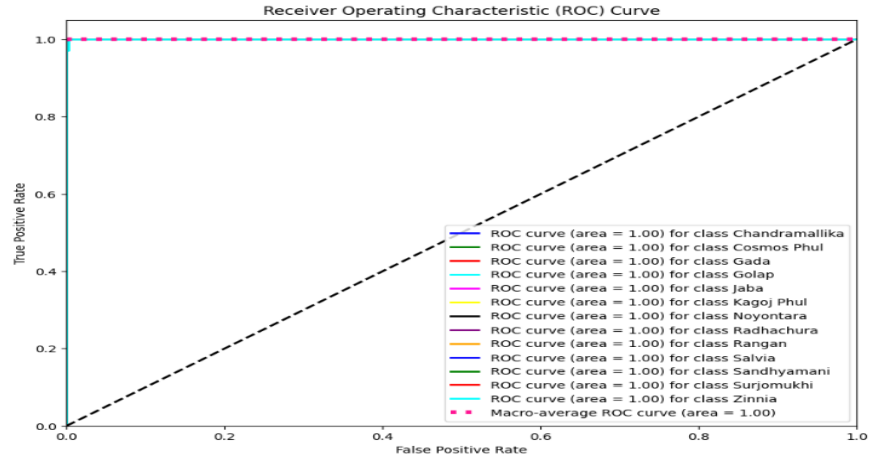


Figure 5.17: ROC Curve of Xception for Single Flowers

❖ ROC curve for Bulk flowers:

1. MobileNetV2:

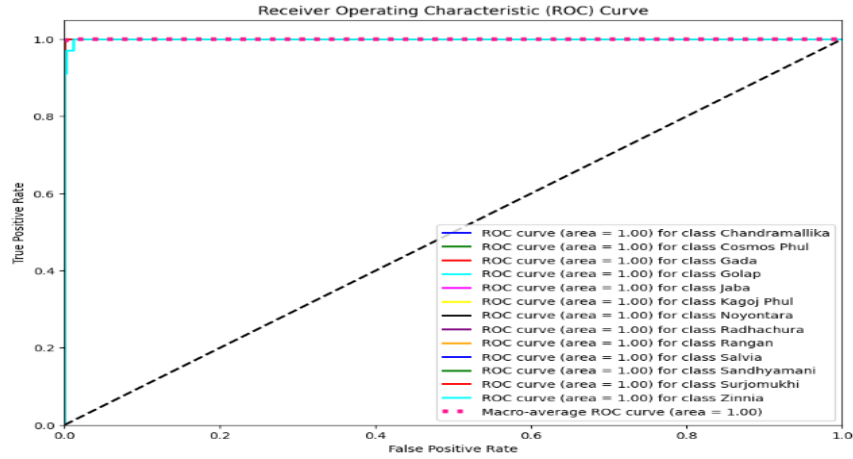


Figure 5.18: ROC Curve of MobileNetV2 for Bulk Flowers

2. EfficientNetV2B0:

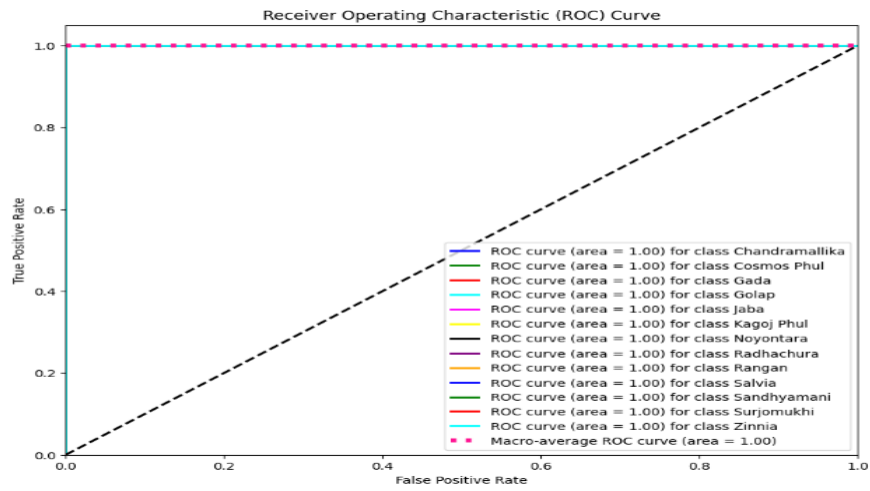


Figure 5.19: ROC Curve of EfficientNetV2B0 for Bulk Flowers

3. ResNet50V2:

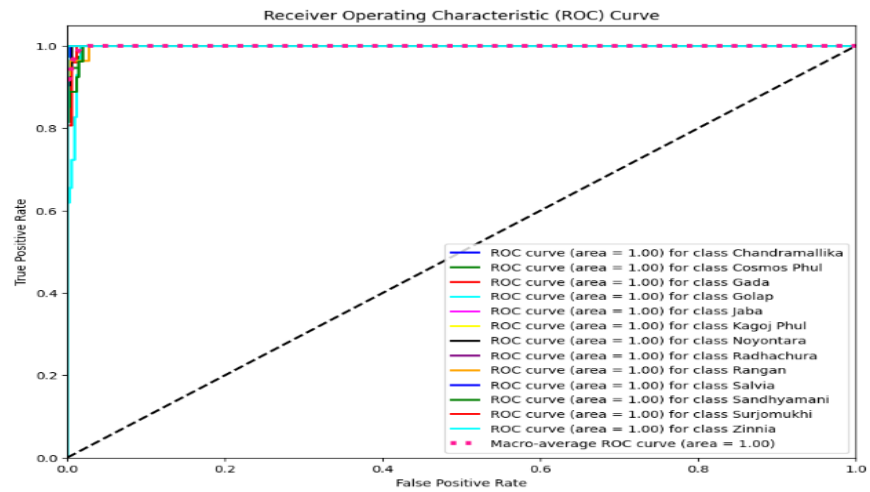


Figure 5.20: ROC Curve of ResNet50V2 for Bulk Flowers

4. InceptionV3:

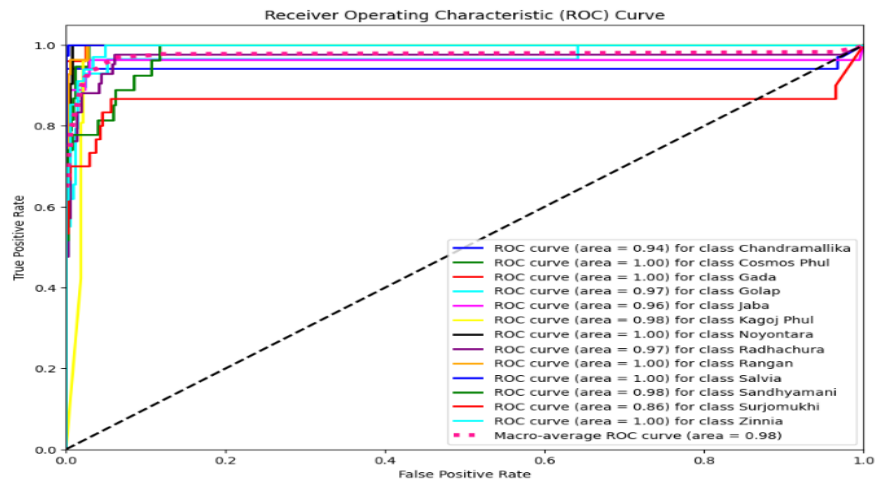


Figure 5.21: ROC Curve of InceptionV3 for Bulk Flowers

5. Xception:

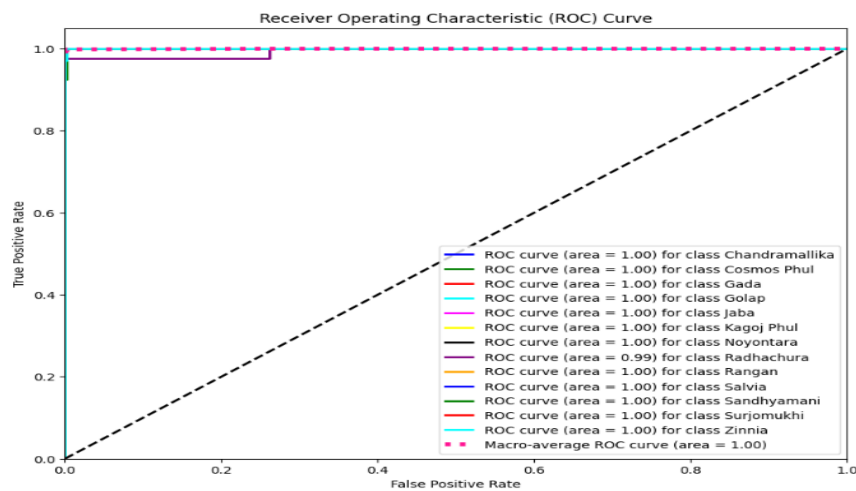


Figure 5.22: ROC Curve of Xception for Bulk Flowers

Observations:

- The ROC curve lies well above the diagonal, indicating strong discriminative performance.
- The AUC score is approximately 1.00, showing the model's high classification ability.

5.2.3 Confusion Matrix

The confusion matrix presents a summary of prediction results, showing the counts of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). This helps evaluate accuracy and class-wise performance. Accuracy alone is not enough to assess a model's overall performance. Each model's precision, recall, and f1-score have been addressed before

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{F1-Score} = 2 \times (\text{precision} \times \text{recall}) / (\text{precision} + \text{recall})$$

here,

TP = True Positives,

TN = True Negatives,

FP = False Positives,

FN = False Negatives

❖ Confusion Matrix for Single flowers:

1. MobileNetV2:

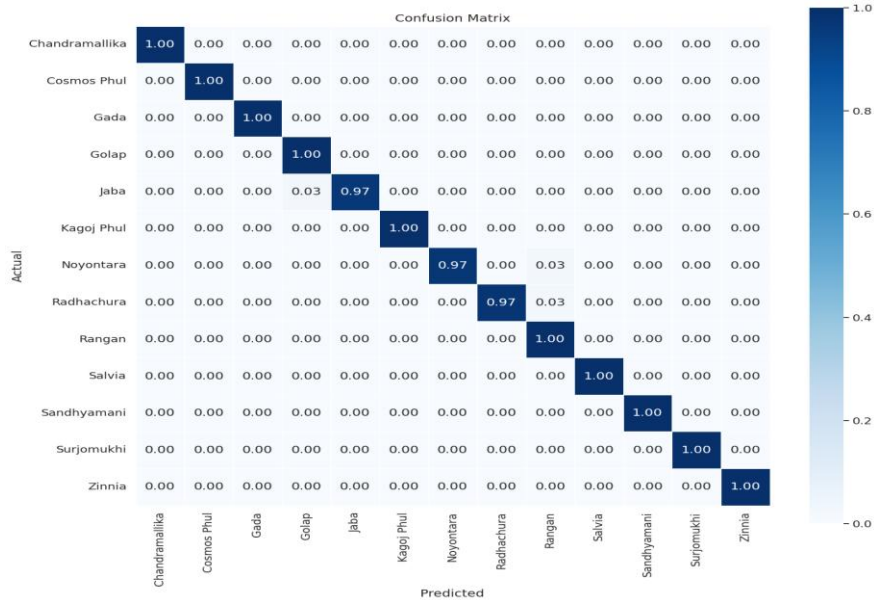


Figure 5.23: Confusion Matrix of MobileNetV2 for Single Flowers

2. EfficientNetV2B0:

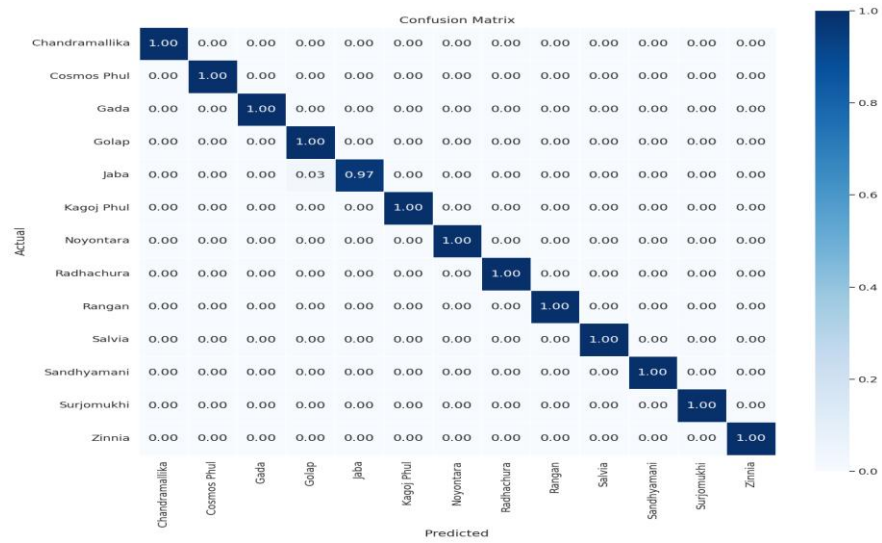


Figure 5.24: Confusion Matrix of EfficientNetV2B0 for Single Flowers

3. ResNet50V2:

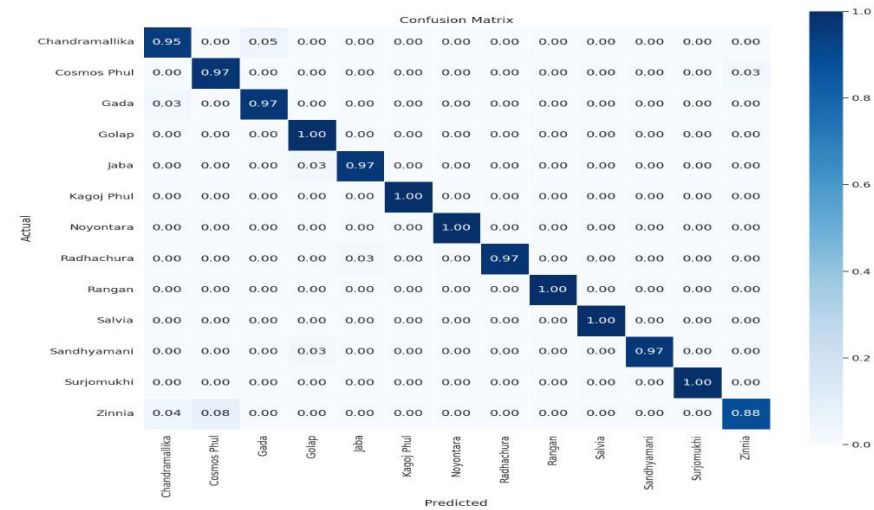


Figure 5.25: Confusion Matrix of ResNet50V2 for Single Flowers

4. InceptionV3:

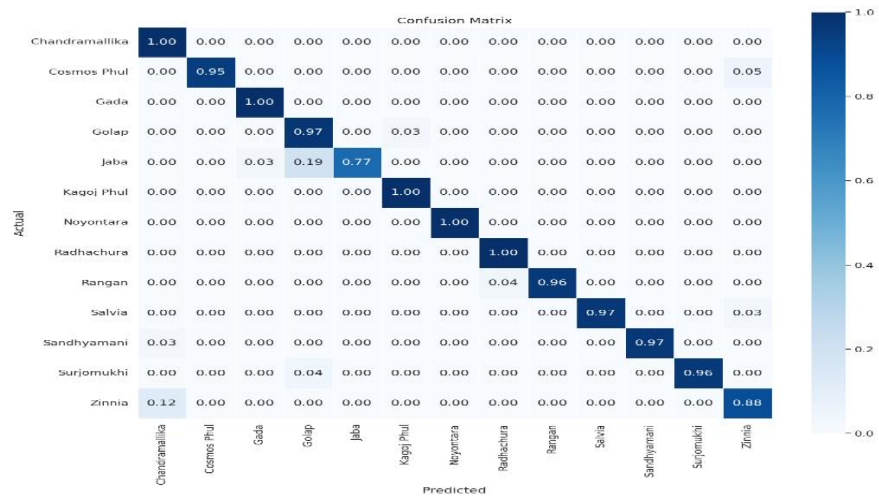


Figure 5.26: Confusion Matrix of InceptionV3 for Single Flowers

5. Xception:

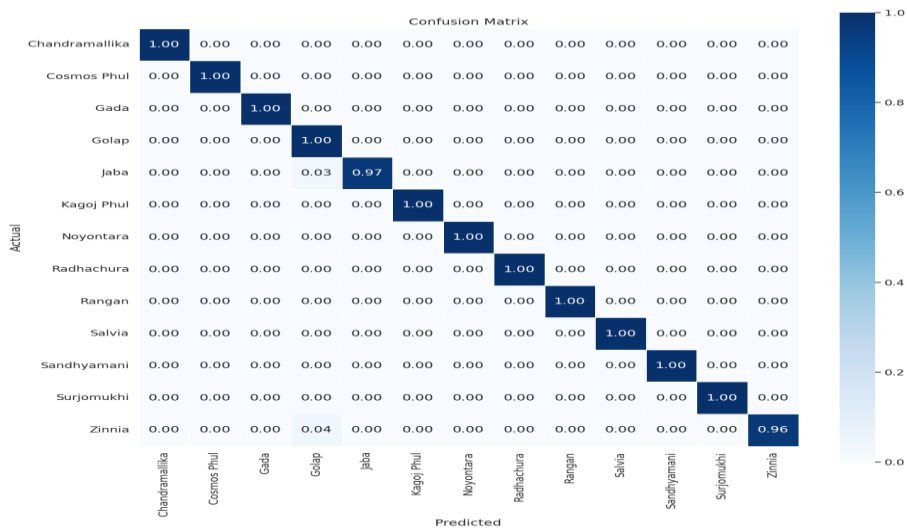


Figure 5.27: Confusion Matrix of Xception for Single Flowers

❖ Confusion Matrix for Bulk Flowers:

1.MobileNetV2:

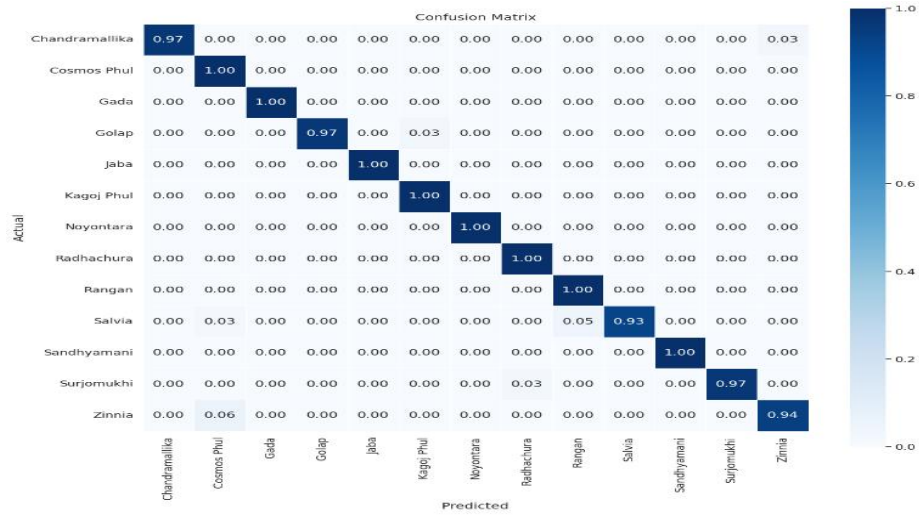


Figure 5.28: Confusion Matrix of MobileNetV2 for Bulk Flowers

2.EfficientNetV2B0:

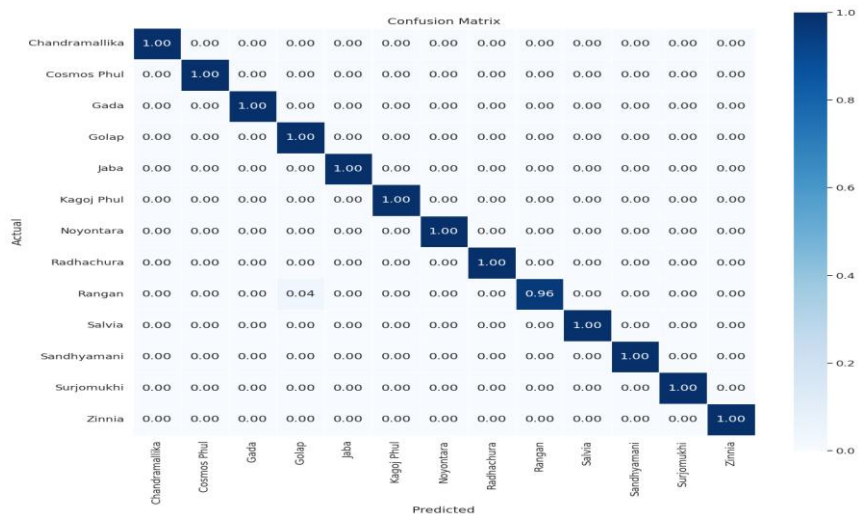


Figure 5.29: Confusion Matrix of EfficientNetV2B0 for Bulk Flowers

3. ResNet50V2:

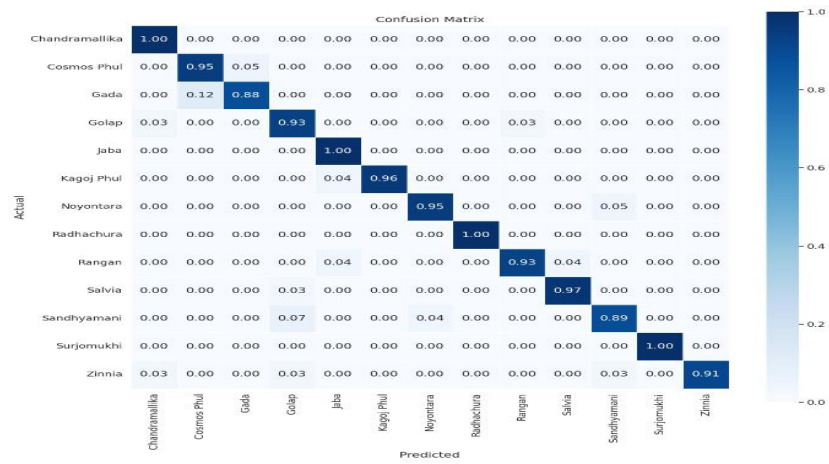


Figure 5.30: Confusion Matrix of ResNet50V2 for Bulk Flowers

4. InceptionV3:

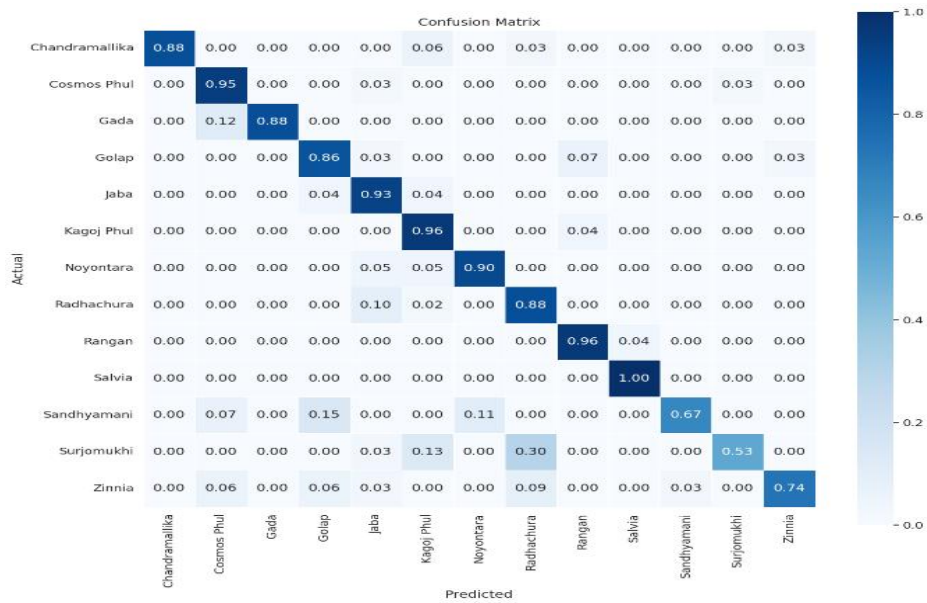


Figure 5.31: Confusion Matrix of InceptionV3 for Bulk Flowers

5. Xception:

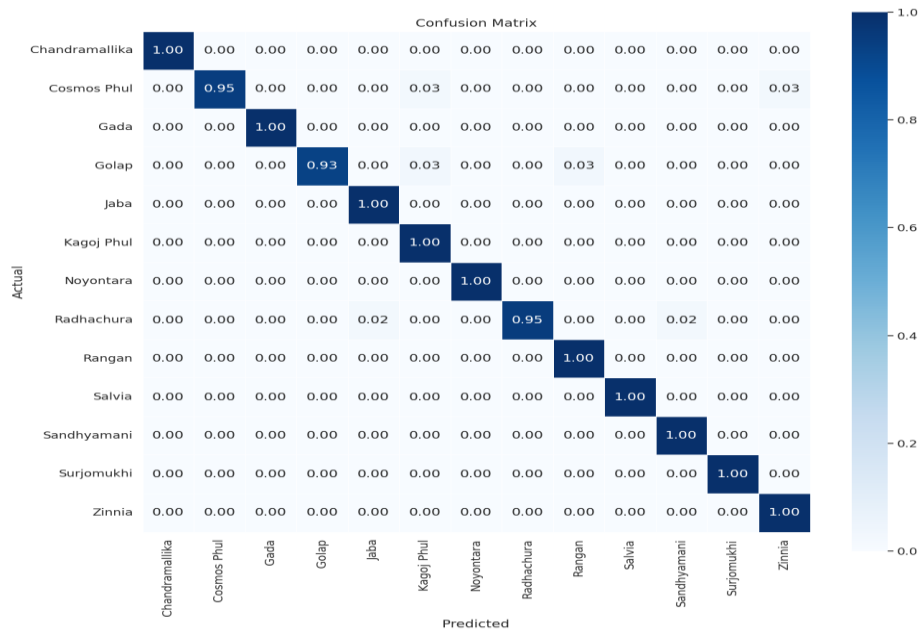


Figure 5.32: Confusion Matrix of Xception for Bulk Flowers

Observations:

- The model accurately classifies most instances.
- Misclassifications are minimal and mostly occur between visually similar categories.
- Class-wise precision and recall scores indicate balanced performance across both classes.

5.3 Discussion:

In this section, we present a comprehensive analysis divided into two parts: Model Performance Analysis and In-Depth Understanding through Curves and Matrices are given below:

Model Performance Analysis: The evaluation of five deep learning models, including MobileNetV2, EfficientNetV2B0, ResNet50V2, InceptionV3, and Xception, reveals varying levels of performance in Bangladeshi local-colored flowers recognition. Each model demonstrates unique strengths and weaknesses in terms of precision, recall, F1-score, and overall accuracy. Notably, EfficientNetV2B0 stands out with exceptional performance, achieving an impressive overall accuracy of 99.75% in both single and bulk category. This underscores the importance of thoughtful model selection based on specific task requirements, considering factors such as architecture, computational efficiency, and overall performance.

In-Depth Understanding through Curves and Matrices: The in-depth analysis of loss and accuracy curves for each model, such as MobileNetV2, EfficientNetV2B0, ResNet50V2, InceptionV3, and Xception, provides valuable insights into their training dynamics. The examination of classification reports and confusion matrices further enhances our understanding of each model's proficiency in distinguishing between different classes. For instance, EfficientNetV2B0 exhibits not only high accuracy but also remarkable precision, recall, and F1-score across various classes. This level of detail allows for a nuanced understanding of the models' capabilities and aids in informed decision-making for specific image classification tasks.

5.4 Summary

The comprehensive evaluation of five deep learning models for flower identification, namely MobileNetV2, EfficientNetV2B0, ResNet50V2, InceptionV3, and Xception, highlights the diverse performance metrics and characteristics of each model. EfficientNetV2B0 emerges as a top performer, showcasing exceptional accuracy and precision. The examination of loss and accuracy curves provides insights into the training dynamics, revealing nuances such as convergence speed and potential overfitting tendencies. The detailed analysis of classification reports and confusion matrices offers a granular understanding of each model's ability to classify instances across multiple classes. In summary, the findings emphasize the importance of tailored model selection based on specific task requirements and highlight the need for a nuanced understanding of each model's strengths and weakness.

Chapter 6

Mobile App Development

Our mobile application named “FloraLens” designed for real-time flower species identification. By leveraging a device's camera or uploaded images, the app provides users with the common and scientific names of flowers. It functions through a client-server model, where the intuitive mobile interface communicates with a powerful backend API built with Python and Flask to perform the classification. Here is the workflow of our mobile app named “FloraLens”.

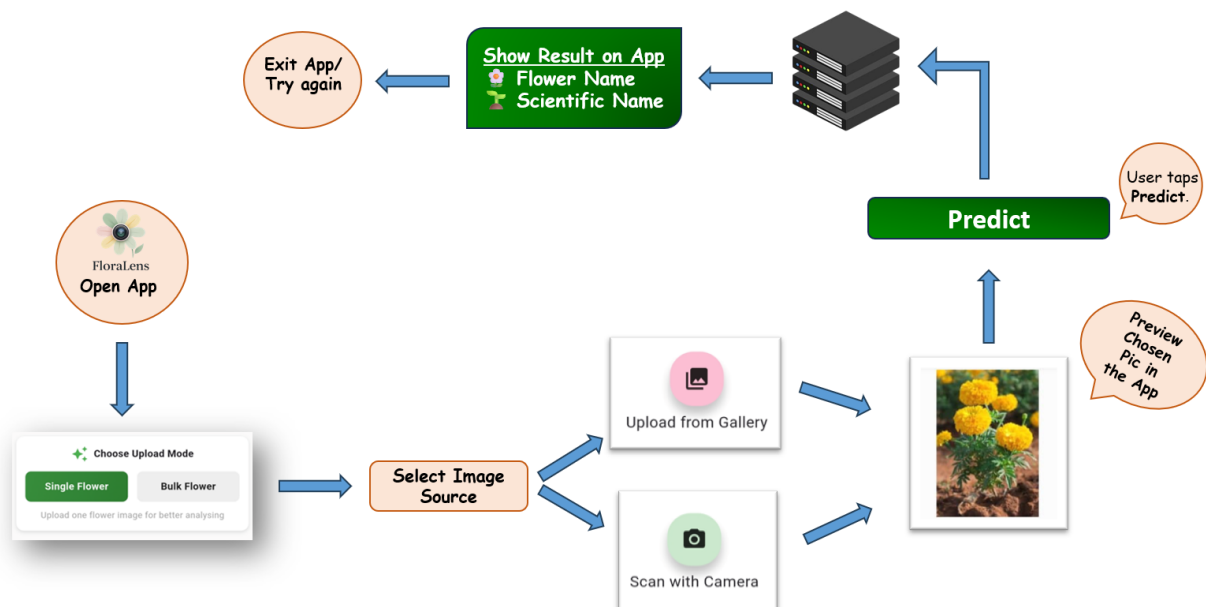


Figure 6.1: Workflow of Mobile Application (FloraLens)

6.1 Mobile App Development Specifications

1. Frontend:

Framework: Flutter with Dart

Architecture: MVVC (Model-View-View Controller) pattern

State Management: GetX for efficient reactive state handling

Networking: HTTP package for API communication

2. Backend:

- Framework: RESTful API built with Flask on Python 3.11

- API Connectivity: Ngrok tunneling service used to generate a public API Base URL, enabling the Flutter app to connect with the locally hosted backend during development and testing

Server Hosting

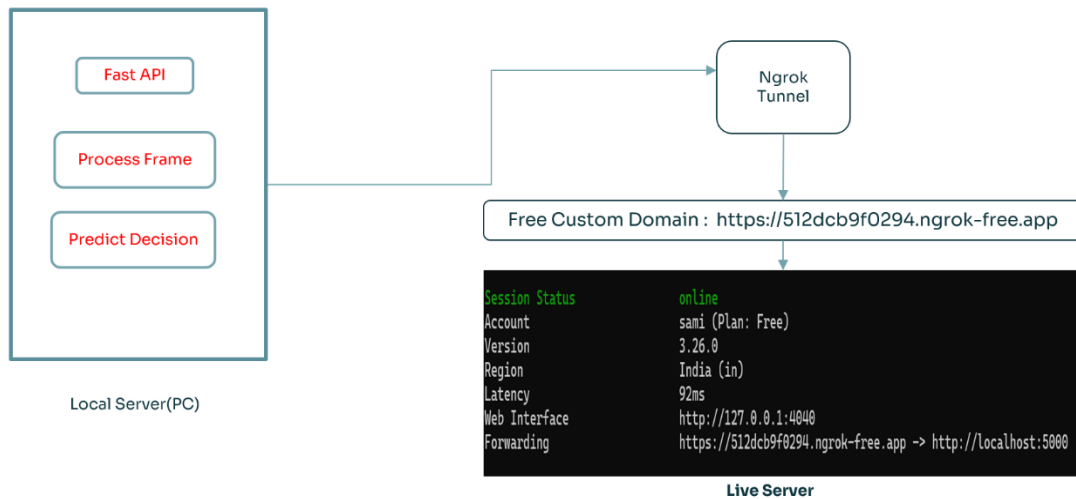


Figure 6.2: Architecture for Local Server Hosting

Development Environment Specification

- IDE: Android Studio Narwhal | 2025.1.1
- Build System: Gradle 8.2
- Android Configuration: SDK 21
- Testing Emulator: Pixel 4 API 33
- VS Code: For Flask API integration with the ML model

6.2 User Interface & Experience

"FloraLens" features a clean, modern, and intuitive interface designed for smooth interaction. The workflow is simple and efficient, ensuring users can easily navigate without any prior technical knowledge. Clear icons and minimal design elements reduce distractions and make the identification process straightforward. The app focuses on usability, so each step feels natural, from capturing or uploading a flower image to instantly viewing the result. The interface is lightweight and responsive, allowing seamless use across different devices.

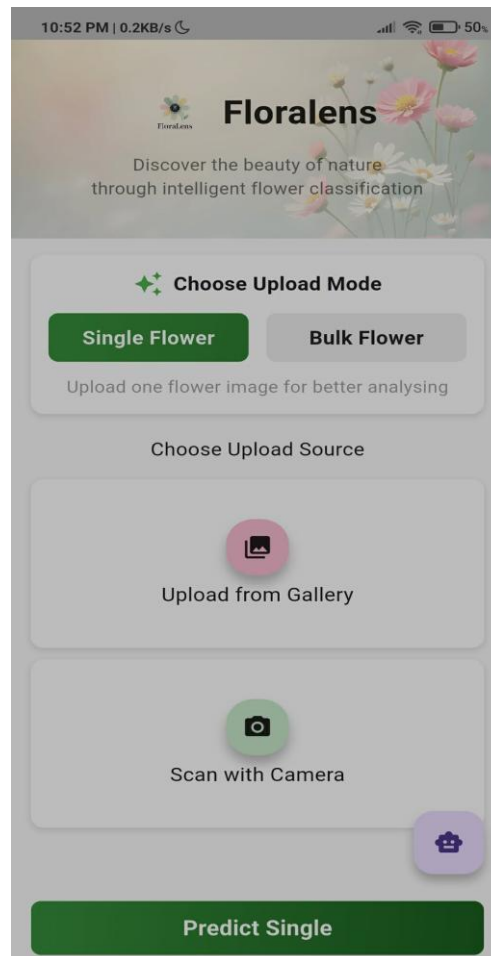


Figure 6.3: UI of Mobile Applications

1. Upload Mode:

Provides two options for selecting images from the device:

- **Single Flower Mode:**
 - Upload one image containing a single flower.
 - The app analyzes the image and shows the common name and scientific name of the flower.
 - Ideal for quick and focused identification.
- **Bulk Flower Mode**
 - Upload one image that contains multiple flowers.
 - The system detects each flower in the image, processes them, and displays the name and scientific name for every flower found.
 - Useful for analyzing images taken in gardens, research plots, or field surveys where many flowers appear together.

2. Capture Mode:

Uses the device camera for Real-time recognition:

- Point the camera at a flower and capture the image.
- The app immediately identifies the flower and displays the name and scientific name on screen.

Real time System integration with Android App

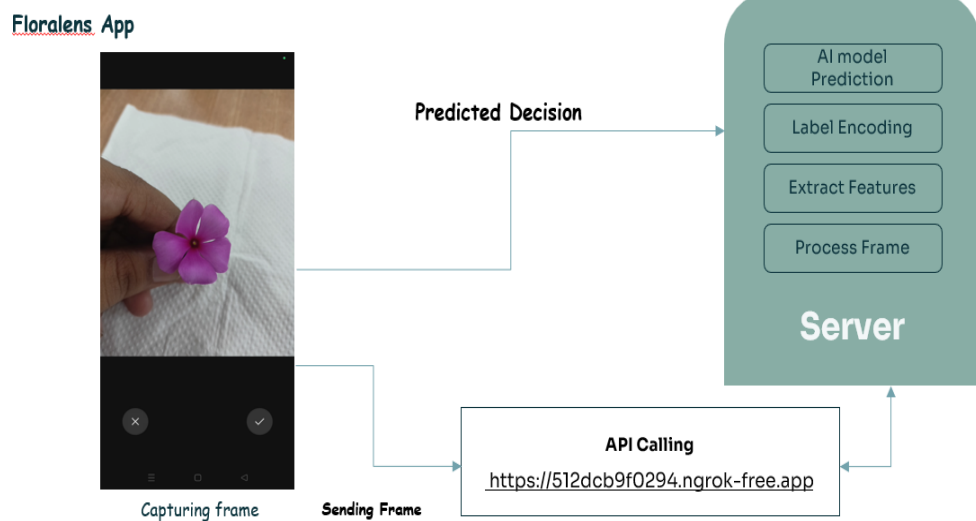


Figure 6.4: Client-Server Architecture for Real-time Image Recognition

- Results: Displayed in a minimalistic pop-up, showing both common and scientific names without clutter.

Screenshots demonstrate accurate recognition of species such as Zinnia and KagojPhul (Bougainvillea) along with their scientific names.

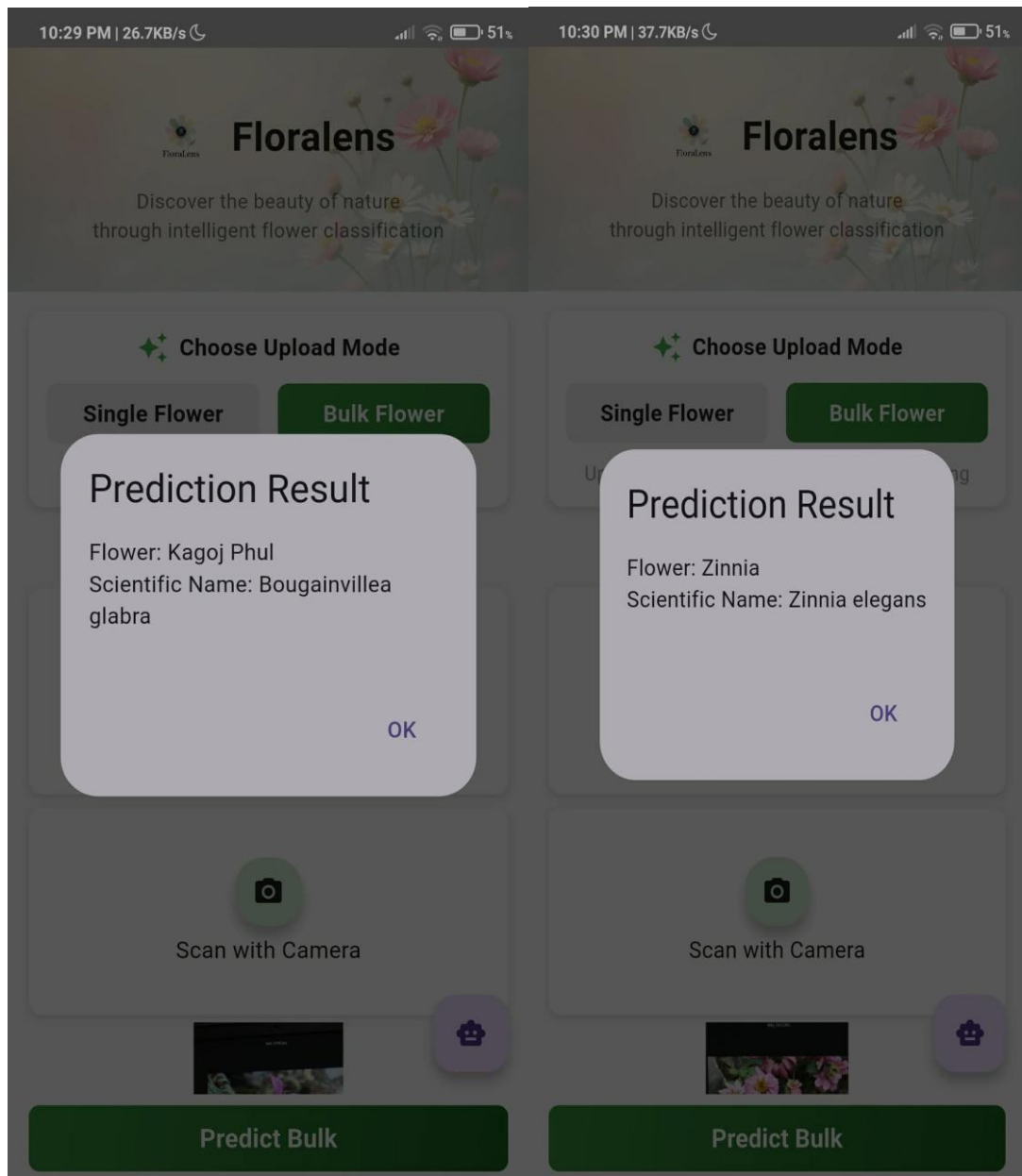


Figure 6.5: Sample Prediction of Flowers with Scientific Name

Chapter 7

Conclusion

This research successfully demonstrates a CNN-based computer vision approach for the accurate recognition and classification of 13 distinct local colored flower varieties found in Bangladesh. Using a high-quality, well-balanced dataset and leveraging five pre-trained deep learning models—MobileNetV2, EfficientNetV2B0, ResNet50V2, InceptionV3, and Xception—the study provides a comparative analysis of their performance in terms of accuracy, precision, recall, and F1-score.

Among these, the EfficientNetV2B0 model outperformed all others, achieving an exceptional accuracy of 99.75% in both single and bulk flower classification tasks, making it the optimal choice for practical implementation. The robust training process, effective data augmentation, and extensive evaluation confirm the reliability and scalability of this approach.

Beyond model accuracy, the broader significance of this research lies in its potential applications. This work contributes to the development of automated flower identification systems that can be applied across multiple domains, including computer vision, botanical research, floral biodiversity monitoring, agriculture and horticulture, environmental conservation, AI-driven recognition tools, educational platforms, the food and cosmetics industries, pollination and ecology studies, and aesthetic or design-oriented applications.

To maximize accessibility and practical impact, the trained model has been successfully deployed as mobile application. These platforms provide real-time flower recognition through a clean, intuitive interface, enabling researchers, educators, and general users to identify flowers instantly. By extending availability across devices, the system ensures broader usability and supports a wide range of scientific, educational, and commercial activities related to floral biodiversity in Bangladesh.

Moreover, we are working on Data Article on our collected dataset. It will be submitted in “Data in Brief” which will open new doors for our paper and dataset for further research and development or it would help others to take ideas to do more updated research.

In summary, the study not only achieves high-performance flower classification but also lays the groundwork for scalable, user-friendly AI applications that support conservation, education, and innovation across diverse sectors.

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